

Math 121: Commutative Algebra

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Worksheet 1.1: Basics of Rings

Problem 1. Let R be a ring and consider $R[[t]]$ the ring of formal power series with coefficients in R . The goal of this exercise is prove that:

$$(R[[t]])^\times = \{u + tf(t) \mid u \in R^\times, f(t) \in R[[t]]\}.$$

- (a) Elements of $R[[t]]$ can be written as $f = \sum_{k=0}^{\infty} f_k t^k$ for $f_k \in R$. We call f_0 the constant (or lead) term of f . If $f = \sum_{k=0}^{\infty} f_k t^k$ and $g = \sum_{k=0}^{\infty} g_k t^k$ then find a formula for the lead term of fg .
- (b) Using the formula from (a) prove that if f is a unit then $f_0 \in R^\times$.
- (c) Find a formula for the coefficient of t^k appearing in fg .
- (d) Assume that $f_0 \in R^\times$. Prove that f is a unit by explicitly constructing an element $g \in R[[t]]$ such that $fg = 1$. (Hint: the condition that $fg = 1$ places constraints on each coefficient of fg . Use the constraints together with part (c) to recursively solve for the coefficients of g .)

Solution.

- (a) Formula for the constant term of fg is $f_0 g_0$.
- (b) Assume f is a unit. Then there exists some $f' \in R[[t]]$ such that $ff' = 1$. By part (1), the constant term of ff' is $f_0 f'_0$. The constant term of 1 is 1 , so $f_0 f'_0 = 1$. Hence f_0 is a unit, and $f_0 \in R^\times$.
- (c) The formula for the coefficient of t^k is $\sum_{i=0}^k f_i g_{k-i}$.
- (d) Assume $f_0 \in R^\times$. Let its inverse be g_0 . For f to be a unit, every non-leading coefficient of fg must be 0 . That is,

$$fg = f_0 g_0 + \sum_{k=1}^{\infty} \left(\sum_{i=0}^k f_i g_{k-i} \right) t^k$$

where $\sum_{i=0}^k f_i g_{k-i} = 0$ for all k . Then solving for g_k gives us,

$$g_k = -\frac{1}{f_0} \sum_{i=0}^{k-1} f_i g_{k-i} = -g_0 \sum_{i=0}^{k-1} f_i g_{k-i}$$

Problem 2.

- (a) Prove that an element $u \in R$ is a unit if and only if the principal ideal $\langle u \rangle = R$.
- (b) Prove that if ab and b are units, then a is a unit.
- (c) Prove that if u is a unit and $a \in R$, then a and ua generate the same principal ideal.
- (d) In the ring $\mathbb{Z}/\langle 12 \rangle$, determine which elements are units.

Solution.

- (a) ($\implies$) Assume $u \in R$ is a unit.

Then there exists some $u' \in R$ such that $uu' = 1 \in \langle u \rangle$. If 1 is in the ideal generated by u then $\forall r \in R, r \in \langle u \rangle$. Hence $\langle u \rangle = R$.

($\impliedby$) Assume $\langle u \rangle = R$ for some $u \in R$.

Then $1 \in \langle u \rangle$. That is, by definition of an ideal, 1 is a multiple of u . Hence, there exists some $u' \in R$ such that $uu' = 1$. Hence, u is a unit.

- (b) Let $a, b \in R$, and ab, b be units. Then there exist $x, b^{-1} \in R$ such that

$$abx = 1, \quad bb^{-1} = 1$$

Then we have,

$$abx = bb^{-1} \implies \frac{abx}{bb^{-1}} = 1 \implies a \cdot \frac{x}{b^{-1}} = 1$$

Hence, a is a unit and $\frac{x}{b^{-1}}$ is its multiplicative inverse.

- (c) Let $u \in R^\times, a \in R$.

Note that the containment $\langle ua \rangle \subseteq \langle a \rangle$ holds by definition. So we only need to show that $\langle a \rangle \subseteq \langle ua \rangle$. Since u is a unit, there exists some $u^{-1} \in R$ such that $uu^{-1} = 1$. Then, note that $uu^{-1}a = a \in \langle ua \rangle$, i.e. $\langle a \rangle \subseteq \langle ua \rangle$.

Hence a and ua generate the same principal ideal.

- (d) $(\mathbb{Z}/\langle 12 \rangle)^\times = \{1, 5, 7, 11\}$

Problem 3. Given two rings $\mathbf{R}$ and $\mathbf{S}$ we let $\text{Hom}_{\text{Ring}}(\mathbf{R}, \mathbf{S})$ be the set of ring homomorphism from $\mathbf{R}$ to $\mathbf{S}$. When there is little room for confusion we will often drop the subscript Ring and simply write $\text{Hom}(\mathbf{R}, \mathbf{S})$.

- (a) Prove or disprove: If $\mathbf{R}$ and $\mathbf{S}$ are rings, then $\text{Hom}(\mathbf{R}, \mathbf{S})$ is a ring under point-wise multiplication and point-wise addition.
- (b) Give an explicit description of the hom-sets below:
 - (a) $\text{Hom}(\mathbb{Z}, \mathbb{Z})$
 - (b) $\text{Hom}(\mathbb{Z}/\langle 2 \rangle, \mathbb{Z})$
 - (c) $\text{Hom}(\mathbb{Z}, \mathbb{Z}/\langle 2 \rangle)$
 - (d) $\text{Hom}(\mathbb{Z}, \mathbb{C}[x])$
- (c) Based on part (b) state and prove a characterization of $\text{Hom}(\mathbb{Z}, \mathbf{R})$ for any ring $\mathbf{R}$.

Solution.

- (a) Let $f, g \in \text{Hom}(\mathbf{R}, \mathbf{S})$. Note that since ring homomorphisms are unital $f(1_{\mathbf{R}}) = 1_{\mathbf{S}}$ and $g(1_{\mathbf{R}}) = 1_{\mathbf{S}}$. Assume for the sake of contradiction that $\text{Hom}(\mathbf{R}, \mathbf{S})$ is a ring under pointwise addition. Then,

$$1 = (f + g)(1) = f(1_{\mathbf{R}}) + g(1_{\mathbf{R}}) = 1 + 1 = 2$$

A contradiction! Hence $\text{Hom}(\mathbf{R}, \mathbf{S})$ is not a ring under pointwise addition and pointwise multiplication.

- (b)
 - (i) $\text{Hom}(\mathbb{Z}, \mathbb{Z}) = \{\text{id}\}$
 - (ii) $\text{Hom}(\mathbb{Z}/\langle 2 \rangle, \mathbb{Z}) = \emptyset$
 - (iii) $\text{Hom}(\mathbb{Z}, \mathbb{Z}/\langle 2 \rangle) = \{\mathbb{Z} \twoheadrightarrow \mathbb{Z}/\langle 2 \rangle\}$
 - (iv) $\text{Hom}(\mathbb{Z}, \mathbb{C}[x]) = \{\mathbb{Z} \hookrightarrow \mathbb{C}[x]\}$
- (c) Note that since both $\mathbb{Z}$ and $\mathbf{R}$ are rings, any homomorphism $\phi : \mathbb{Z} \rightarrow \mathbf{R}$ is unital, i.e. $\tau(1) = 1_{\mathbf{R}}$. To be a homomorphism we must have $\forall a \in \mathbb{Z}$,

$$\tau(a) = \tau\left(\sum_{i=1}^a 1\right) = \sum_{i=1}^a \tau(1)$$

This completely determines the homomorphism, and thus there is a unique (unital) homomorphism from $\mathbb{Z}$ to any ring $\mathbf{R}$.

Problem 4. Let $\phi : R \rightarrow S$ be a ring homomorphism.

- (a) Prove that if $J \subset S$ is an ideal then $\phi^{-1}(J) \subset R$ is an ideal.
- (b) Assuming ϕ is injective, give a nice description of $\phi^{-1}(J)$ for an ideal $J \subset S$.
- (c) Find a counterexample showing that if $I \subset R$ is an ideal $\phi(I)$ need not be an ideal.
- (d) Prove that if ϕ is surjective then $\phi(I)$ is an ideal of S .
- (e) Let $I \subset R$ be an ideal. Prove that if $\ker(\phi) \subset I$ then $\phi^{-1}(\phi(I)) = I$.
- (f) Is the assumption that $\ker(\phi) \subset I$ needed in part (5)?

Solution.

- (a) Let $a, b \in \phi^{-1}(J)$. Then $\phi(a), \phi(b) \in J$. Since ideals are closed under addition, $\phi(a) + \phi(b) = \phi(a + b) \in J$. Thus $\phi^{-1}(a + b) \in \phi^{-1}(J)$.
Similarly let $r \in R$, and $a \in \phi^{-1}(J)$. Then $\phi(r) \cdot \phi(a) = \phi(r \cdot a) \in J$. Thus, $\phi^{-1}(r \cdot a) \in \phi^{-1}(J)$.

- (b) If ϕ is injective then,

$$\phi^{-1}(J) = \{r \in R \mid \phi(r) \in J\}$$

$$\phi^{-1}(J) = J \cap \phi(R) = J \cap R$$

- (c) Let $R = \mathbb{Z}$, $S = \mathbb{Z}[x]$ and define the homomorphism to be $\phi : R \hookrightarrow S$. Then consider the image of the ideal generated by 3. It does not contain $3x$ and hence is not closed under multiplication.

- (d) Let $I \subset R$ be an ideal and assume that ϕ is surjective. Let $a, b \in I$. We want to show that $\phi(I)$ is a subgroup of the additive group of S and is closed under multiplication by elements of S .

Consider $\phi(a) - \phi(b)$. Note that since ϕ is a homomorphism the following holds,

$$\phi(a) - \phi(b) = \phi(a - b)$$

and since $a - b \in I$, $\phi(a - b) \in \phi(I)$. Moreover since $\phi(0) = 0 \in \phi(I)$, $\phi(I)$ is a subgroup of S under addition.

Now note that since ϕ is surjective, $\forall s \in S \exists r \in R$ such that $s = \phi(r)$. Then since $a \in I$, $r \in R$ and I is an ideal of R we have $ra \in I$. So,

$$s \cdot \phi(a) = \phi(r)\phi(a) = \phi(ra) \in \phi(I)$$

Hence, $\phi(I)$ is closed under multiplication by elements of S and thus is an ideal of S .

(e) Assume $\ker(\phi) \subset I$.

$$I \subset \phi^{-1}(\phi(I))$$

Since $\phi^{-1}(\phi(I))$ is the pre-image of $\phi(I)$, this containment is straightforward.

$$\phi^{-1}(\phi(I)) \subseteq I$$

Let $x \in \phi^{-1}(\phi(I))$. That is $\phi(x) = \phi(a)$ for some $a \in I$.

$$\phi(x - a) = 0 \implies x - a \in \ker(\phi) \subset I$$

Since I is closed under addition $x \in I$. Hence, $\phi^{-1}(\phi(I)) \subseteq I$

(f) Yes. As a counterexample, consider the quotient map $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/\langle 3 \rangle$ and the ideal generated by 2, $I = \langle 2 \rangle$. Note that $0 \in \phi(I) \implies \ker(\phi) = 3\mathbb{Z} \subset \phi^{-1}(\phi(I))$. That is $3 \in \phi^{-1}(\phi(I))$ but $3 \notin I$.

Problem 1. Operations on Ideals.

Throughout this problem let I and J be ideals in a ring R .

- (a) Prove or disprove that $I \cap J$ is an ideal.
- (b) Prove or disprove that $I \cup J$ is an ideal.
- (c) Prove or disprove that the product of ideals, defined below, is an ideal:

$$IJ := \{a_1b_1 + \cdots + a_tb_t \mid a_i \in I, b_i \in J, \text{ for all } t\}.$$

- (d) Prove or disprove that the sum of ideals, defined below, is an ideal:

$$I + J := \{a + b \mid a \in I, b \in J\}.$$

- (e) Prove or disprove that the radical of an ideal, defined below, is an ideal:

$$\sqrt{I} := \{a \in R \mid a^k \in I \text{ for some } k \in \mathbb{Z}_{\geq 1}\}.$$

Solution.

- (a) $I \cap J$ is an ideal.

Let $a, b \in I \cap J$, $r \in R$.

- (i) $I \cap J$ is a subgroup because intersection of subgroups is a subgroup.
- (ii) Note that $a, b \in I, J$. Since I and J are closed under addition and inverses, $a - b \in I, J$. Hence $a - b \in I \cap J$.
- (iii) $r \cdot a \in I \cap J$ since $ra \in I$ and $ra \in J$.

Hence $I \cap J$ is an additive subgroup of R and is closed under multiplication from elements of R .

- (b) $I \cup J$ is not an ideal.

Consider the ideals $I = \langle 2 \rangle$, $J = \langle 3 \rangle$ in the ring $R = \mathbb{Z}$. Note that $5 \notin I \cup J$ even though $2, 3 \in I \cup J$. Hence, $I \cup J$ is not closed under addition and is not an ideal.

- (c) IJ is an ideal.

Let $s \in IJ$, $r \in R$.

- (i) By definition, IJ includes all finite sums of elements in I, J . So it contains 0 and is closed under addition and inverses. Hence it is an additive subgroup.
- (ii) Consider $r \cdot s \in IJ$.

$$r \cdot s = r(a_1 b_1 + \cdots + a_t b_t) = r \cdot a_1 b_1 + \cdots + r \cdot a_t b_t$$

Note that $r \cdot a_i \in I$ for all i and thus $rs \in IJ$ for all r .

Hence IJ is an additive subgroup of R and is closed under multiplication from elements of R .

- (d) $I + J$ is an ideal.

Let $p = a_1 + b_1, q = a_2 + b_2 \in I + J, r \in R$.

- (i) Note that $0 \in I + J$ since $0 \in I, J$.
- (ii) Consider $p - q$.

$$p - q = a_1 + b_1 - (a_2 + b_2) = (a_1 - a_2) + (b_1 - b_2)$$

Note that $(a_1 - a_2) = a$ for some $a \in I$ and $(b_1 - b_2) = b$ for some $b \in I$. Hence, $I + J$ contains zero and is closed under addition and inverses and thus is an additive subgroup.

- (iii) Consider $r \cdot p$.

$$r \cdot p = r \cdot (a_1 + b_1) = ra_1 + rb_1$$

Note that $ra_1 \in I$ and $rb_1 \in J$ by definition of ideals. Hence $rp \in I + J$.

Thus $I + J$ is an additive subgroup of R and is closed under multiplication from elements of R .

- (e) $\sqrt{I}$ is an ideal.

Let $a^n, b^m \in I$, and $a, b, r \in R$.

- (i) Note that $0 \in \sqrt{I}$ since $0^1 \in I$.
- (ii) Consider $(a - b)^{nm}$.

$$(a - b)^{nm} = \sum_{i=0}^{nm} \binom{nm}{i} (a)^i (-b)^{nm-i}$$

- If $i \geq n$, then $a^n \in I \implies \binom{nm}{i} (a)^i (-b)^{nm-i} \in I$.
- If $i < n$, then $nm - i > nm - n = n(m - 1) = nm - i \geq m$. And $b^m \in I \implies \binom{nm}{i} (a)^i (-b)^{nm-i} \in I$.

Since every term in the sum lies in I , the whole sum lies in I . So $a - b \in \sqrt{I}$.

- (iii) Note that if $a^n \in I$ then $(ra)^n \in I$. Hence $ra \in \sqrt{I}$.

Thus $\sqrt{I}$ is an additive subgroup of R and is closed under multiplication from elements of R .

Problem 2. Let $\mathbb{K}$ be a field. The rings $\mathbb{Z}$, $\mathbb{K}[x]$, and $\mathbb{K}[[t]]$ are all principal ideal domains (PIDs) meaning domains where every ideal is generated by a single element. If $I = \langle a \rangle$ and $J = \langle b \rangle$ give, with proof, a careful concrete description of generators for IJ , $I \cap J$, $I + J$, and $\sqrt{I}$ in each of these rings.

Solution.

Let $I = \langle a \rangle$, $J = \langle b \rangle$. Then, we have

(a) $\mathbb{Z}$

- (i) $IJ = \langle ab \rangle$
- (ii) $I \cap J = \langle \text{lcm}(a, b) \rangle$
- (iii) $I + J = \langle \text{gcd}(a, b) \rangle$
- (iv) Let $a = p_1^{e_1} \cdots p_k^{e_k}$, then $\sqrt{I} = \langle p_1 \cdots p_k \rangle$, where p_i is a prime element.

(b) $\mathbb{K}[x]$

- (i) $IJ = \langle ab \rangle$
- (ii) $I \cap J = \langle \text{lcm}(a, b) \rangle$
- (iii) $I + J = \langle \text{gcd}(a, b) \rangle$
- (iv) Let $a = p_1^{e_1} \cdots p_k^{e_k}$, then $\sqrt{I} = \langle p_1 \cdots p_k \rangle$, where p_i is an irreducible element.

(c) $\mathbb{K}[[t]]$ Note that elements of $\mathbb{K}[[t]]$ are of two forms: $a \in \mathbb{K}$ or $a \cdot t^n$, $a \in \mathbb{K}$. So all proper ideals are of the form $\langle 0 \rangle$ or $\langle t^n \rangle$. Let $I = \langle t^n \rangle$, $J = \langle t^m \rangle$. If I is trivial, then $IJ = I \cap J = \sqrt{I} = \langle 0 \rangle$, and $I + J = J$. Assuming neither I nor J are trivial we have,

- (i) $IJ = \langle t^{n+m} \rangle$
- (ii) $I \cap J = \langle t^{\max(n, m)} \rangle$
- (iii) $I + J = \langle t^{\min(n, m)} \rangle$
- (iv) $\sqrt{I} = \langle t \rangle$.

Universal Properties. Throughout this course we will see numerous constructions and operations (e.g., quotients, tensor products, localizations, completions, etc.) each of these will have an explicit construction where we describe the objects precisely as sets. However, there is an alternative perspective, which is just as useful, describing these objects via their *universal properties*, which roughly asks what makes each object special amongst all other such objects. The standard framework for a universal property is, “Object X is unique up to isomorphism among all objects having property P.”

Problem 3. Universal Property of Quotients.

Let R be a ring and $I \subseteq R$ an ideal. Let $\pi: R \rightarrow R/I$ be the quotient map. Suppose $\varphi: R \rightarrow S$ is a ring homomorphism such that $I \subseteq \ker(\varphi)$. Define a function

$$\tilde{\varphi}: R/I \rightarrow S \quad \text{by} \quad \tilde{\varphi}(r + I) = \varphi(r).$$

- (a) Prove that $\tilde{\varphi}$ is well-defined.
- (b) Prove that $\tilde{\varphi}$ is a ring homomorphism.
- (c) Prove that $\tilde{\varphi} \circ \pi = \varphi$.
- (d) Prove that $\tilde{\varphi}$ is the unique ring homomorphism $R/I \rightarrow S$ with the property that

$$\tilde{\varphi} \circ \pi = \varphi.$$

- (e) Explain in words what this says about the quotient ring R/I .

Solution.

- (a) Let $r + I = r' + I \in R/I$. Then, $r - r' \in I$.

$$\tilde{\varphi}(r + I) = \varphi(r) = \varphi(r + (r - r')) = \varphi(r') = \tilde{\varphi}(r' + I)$$

Hence, $\tilde{\varphi}$ is well-defined.

- (b) Let $\bar{a}, \bar{b} \in R/I$, we check if the three conditions hold.

Unital: Since φ is a ring homomorphism, we have

$$\tilde{\varphi}(1_{R/I}) = \varphi(1_R) = 1_S$$

Addition: Again, since φ is a ring homomorphism, we have

$$\tilde{\varphi}(\bar{a}) + \tilde{\varphi}(\bar{b}) = \varphi(a) + \varphi(b) = \varphi(a + b) = \tilde{\varphi}(\overline{a + b})$$

Multiplication: Again, φ is a ring homomorphism, we have

$$\tilde{\varphi}(\overline{\mathfrak{a}}) \cdot \tilde{\varphi}(\overline{\mathfrak{b}}) = \varphi(\mathfrak{a}) \cdot \varphi(\mathfrak{b}) = \varphi(\mathfrak{a} \cdot \mathfrak{b}) = \tilde{\varphi}(\overline{\mathfrak{a} \cdot \mathfrak{b}})$$

Hence, $\tilde{\varphi}$ is a ring homomorphism.

(c) Let $\mathfrak{a} \in \mathbf{R}$. Then,

$$\tilde{\varphi} \circ \pi(\mathfrak{a}) = \tilde{\varphi}(\overline{\mathfrak{a}}) = \varphi(\mathfrak{a})$$

Since $\mathfrak{a}$ was an arbitrary element in $\mathbf{R}$, this holds for all elements, and hence $\tilde{\varphi} \circ \pi = \varphi$.

(d) Assume there exists some other homomorphism $\sigma : \mathbf{R}/I \rightarrow \mathbf{S}$ such that $\sigma \circ \pi = \varphi$. Then for any $\overline{\mathfrak{a}} \in \mathbf{R}/I$,

$$\sigma(\overline{\mathfrak{a}}) = \sigma \circ \pi(\mathfrak{a}) = \varphi(\mathfrak{a}) = \tilde{\varphi}(\mathfrak{a})$$

(e) Every homomorphism φ such that $I \subseteq \ker(\varphi)$ can be written as $\tilde{\varphi} \circ \pi$ for some $\tilde{\varphi}$.

Problem 4. In this problem, we wish to consider the evaluation map:

$$\text{ev}_i: \mathbb{Z}[x] \rightarrow \mathbb{C}, \quad f(x) \mapsto f(i)$$

- (a) Prove that ev_i is a ring homomorphism.
- (b) Show that $\langle x^2 + 1 \rangle \subset \ker(\text{ev}_i)$.
- (c) Use Problem 3 to produce a ring homomorphism

$$\widetilde{\text{ev}}_i: \mathbb{Z}[x]/\langle x^2 + 1 \rangle \rightarrow \mathbb{C}.$$

- (d) What does $\widetilde{\text{ev}}_i$ do to the class of x in $\mathbb{Z}[x]/\langle x^2 + 1 \rangle$?
- (e) Prove that $\mathbb{Z}[x]/\langle x^2 + 1 \rangle$ is isomorphic to $\mathbb{Z}[i]$. Is $\langle x^2 + 1 \rangle$ prime and/or maximal?

Solution.

- (a) Let $f, g \in \mathbb{Z}[x]$, we check if the three conditions hold.

Unital: Let f_1 be the constant function 1. Then,

$$\text{ev}_i(f_1(x)) = f_1(i) = 1$$

Addition: Since $\mathbb{Z}[x]$ is a ring under pointwise addition and multiplication, we can write,

$$\text{ev}_i(f(x)) + \text{ev}_i(g(x)) = f(i) + g(i) = (f + g)(i) = \text{ev}_i((f + g)(x))$$

Multiplication: Since $\mathbb{Z}[x]$ is a ring under pointwise addition and multiplication, we can write,

$$\text{ev}_i(f(x)) \cdot \text{ev}_i(g(x)) = f(i) \cdot g(i) = (f \cdot g)(i) = \text{ev}_i((f \cdot g)(x))$$

Hence, ev_i is a ring homomorphism.

- (b) Note that $\text{ev}_i(x^2 + 1) = i^2 + 1 = -1 + 1 = 0$. For any $f(x)(x^2 + 1) \in \langle x^2 + 1 \rangle$ we have $\text{ev}_i(f(x)(x^2 + 1)) = f(i) \cdot 0 = 0 \in \ker(\text{ev}_i)$. Thus the ideal generated by $\langle x^2 + 1 \rangle$ under ev_i maps to 0 and hence is a subset of the kernel.

- (c) From Q3 we know that if $\langle x^2 + 1 \rangle \subseteq \ker(\text{ev}_i)$ there exists a unique homomorphism,

$$\widetilde{\text{ev}}_i: \mathbb{Z}[x]/\langle x^2 + 1 \rangle \rightarrow \mathbb{C}, \quad \widetilde{\text{ev}}_i(\overline{f(x)}) = f(i)$$

- (d) $\widetilde{\text{ev}}_i(\overline{x}) = x(i) = i$.

- (e) Define a map $\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}[i]$, $\varphi(f(x)) \rightarrow f(i)$. We will show that $\ker(\varphi) = \langle x^2 + 1 \rangle$.
 $\langle x^2 + 1 \rangle \subseteq \ker(\varphi)$

Note,

$$\varphi(x^2 + 1) = i^2 + 1 = 0 \in \ker(\varphi)$$

$$\ker(\varphi) \subseteq \langle x^2 + 1 \rangle$$

Let $\varphi(f(x)) \in \ker(\varphi)$. If $f(i) = 0$, then $x^2 + 1 \mid f(x)$ in $\mathbb{Z}[x]$ (since $x^2 + 1$ is the minimal polynomial of i over $\mathbb{Q}$ and hence is primitive over $\mathbb{Z}$), and we have,

$$f(x) \in \langle x^2 + 1 \rangle \implies \ker(\varphi) \subseteq \langle x^2 + 1 \rangle$$

Then by the first isomorphism theorem,

$$\mathbb{Z}[x]/\langle x^2 + 1 \rangle \cong \text{im}(\varphi) = \mathbb{Z}[i]$$

Since $\mathbb{Z}[x]$ is an integral domain but not a field, $\langle x^2 + 1 \rangle$ is prime but not maximal.

Problem 5. Universal Property of Polynomial Rings.

The goal of this exercise is to prove the following characterization of polynomial rings. Let $\varphi: R \rightarrow S$ be a ring homomorphism. For any element $s \in S$ there is a unique ring homomorphism $\psi: R[x] \rightarrow S$ such that $\psi(x) = s$ and the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ \downarrow & \nearrow \psi & \\ R[x] & & \end{array} .$$

- (a) Prove there is at most one ring homomorphism $\psi: R[x] \rightarrow S$ with the above properties.
- (b) Construct such a homomorphism explicitly by setting

$$\psi\left(\sum_{i=0}^n a_i x^i\right) = \sum_{i=0}^n \varphi(a_i) s^i.$$

- (c) Conclude that to give a ring homomorphism $R[x] \rightarrow S$ is equivalent to giving:
- a ring homomorphism $R \rightarrow S$, and
 - an element of S .

Solution.

- (a) Suppose there exists another homomorphism $\sigma: R[x] \rightarrow S$ such that $\sigma(x) = \psi(x) = s$. Also, note that since the diagram commutes $\psi|_R = \sigma|_R = \varphi$.

Then for any $\sum_{i=0}^n a_i x^i \in R[x]$ we have,

$$\begin{aligned} \sigma\left(\sum_{i=0}^n a_i x^i\right) &= \sigma(a_1) \cdot \sigma(x) + \cdots + \sigma(a_n) \cdot \sigma(x^n) \\ &= \varphi(a_1) \cdot \psi(x) + \cdots + \varphi(a_n) \cdot \psi(x^n) \\ &= \psi(a_1) \cdot \psi(x) + \cdots + \psi(a_n) \cdot \psi(x^n) \\ &= \psi\left(\sum_{i=0}^n a_i x^i\right) \end{aligned}$$

Hence, $\sigma = \psi$.

(b) Let $\psi : R[x] \rightarrow S$ such that

$$\psi(x) = s, \quad \psi(a) = \varphi(a) \quad \forall a \in R$$

(c) Hence for any ring homomorphism $\sigma : R[x] \rightarrow S$ we have

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ \downarrow & \nearrow \psi & \\ R[x] & & \end{array}$$

Problem 6. Evaluation maps.

Let R be a ring and let $a \in R$. Define the evaluation map

$$\text{ev}_a: R[x] \rightarrow R, \quad f(x) \mapsto f(a).$$

- (a) Prove that ev_a is a surjective ring homomorphism.
- (b) Prove that $\ker(\text{ev}_a) = \langle x - a \rangle$.
- (c) Use Problem 3 to show that ev_a factors through a ring homomorphism

$$R[x]/\langle x - a \rangle \rightarrow R.$$

- (d) Explain what this means if R is a field or R is an integral domain.
- (e) Give an example of a ring R and prime/maximal ideal $I \subset R[x]$ such that $I \neq \ker(\text{ev}_a)$ for all $a \in R$.

Solution.

- (a) For all $r \in R$ consider the constant functions r represented by $f_r \in R[x]$. Then,

$$\text{ev}_a(f_r(x)) = f_r(a) = r$$

- (b) $\ker(\text{ev}_a) \subseteq \langle x - a \rangle$.

Note that for any $p(x) \in \ker(\text{ev}_a)$ we have

$$p(x) = (x - a)q(x) + r(x)$$

for some $q(x), r(x) \in R[x]$. Evaluating at ev_a we have,

$$\text{ev}_a(p(x)) = (a - a)q(a) + r(a) = 0 \implies r(a) = 0$$

That is

$$p(x) = (x - a)q(x) \implies p(x) \in \langle x - a \rangle$$

$$\langle x - a \rangle \subseteq \ker(\text{ev}_a).$$

Let $p \in \langle x - a \rangle$. Since $\langle x - a \rangle$ irreducible we have $p = f(x)(x - a)$, for some $f(x) \in R[x]$,

$$\text{ev}_a(p) = \text{ev}_a(f(x)(x - a)) = f(a)(a - a) = 0 \in \ker(\text{ev}_a)$$

Hence, $\ker(\text{ev}_a) = \langle x - a \rangle$.

- (c) Since $\ker(\text{ev}_a) \supseteq \langle x - a \rangle$ from problem 3 we know we have an induced ring homomorphism $\widetilde{\text{ev}}_a$ such that

$$\begin{array}{ccc} \mathbf{R}[x] & \xrightarrow{\pi} & \mathbf{R}[x]/\langle x - a \rangle \\ \text{ev}_a \downarrow & \swarrow \widetilde{\text{ev}}_a & \\ \mathbf{R} & & \end{array}$$

- (d) If $\mathbf{R}$ is a domain this means $\langle x - a \rangle$ is prime and also that $\widetilde{\text{ev}}_a = \text{id}$.

If $\mathbf{R}$ is a field $\langle x - a \rangle$ is maximal and $\mathbf{R} \cong \mathbf{R}[x]/\langle x - a \rangle$. Also since all domains are fields we have $\widetilde{\text{ev}}_a = \text{id}$

- (e) $\mathbf{R} = \mathbb{C}, I = \langle 0 \rangle$.

Worksheet 1.3: Gröbner Bases Pt. I

Polynomial Division. Recall the following classical theorem:

Theorem 0.1. *Let $g \in \mathbb{K}[x]$ be a non-zero polynomial. Every polynomial $f \in \mathbb{K}[x]$ can be written uniquely as $f = qg + r$ where $q, r \in \mathbb{K}[x]$, and either $r = 0$ or $\deg(r) < \deg(g)$.*

While this theorem might seem innocuous when first learned as a teenager, it turns out to be very powerful when studying ideals and polynomials in $\mathbb{K}[x]$. The proof of this theorem also provides an algorithm for computing q and r , which when presented in pseudocode looks something like

Algorithm 1 Division Algorithm

Require: g, f

Ensure: q, r

$q \leftarrow 0; r \leftarrow f$

while $r \neq 0$ **and** $\text{LT}(g)$ divides $\text{LT}(r)$ **do**

$q \leftarrow q + \text{LT}(r)/\text{LT}(g)$

$r \leftarrow r - (\text{LT}(r)/\text{LT}(g))g$

end while

The goal of this worksheet is to begin exploring how we might extend this theorem to polynomial rings in more variables. Along the way we will see how these explorations will lead to both computational tools for studying the commutative algebra of polynomial rings over fields, but also gives us useful tools for proving theorems in this setting as well.

Problem 1. Use Theorem 1 to prove the following:

- (a) If $f \in \mathbb{K}[x]$ is a non-zero polynomial then f has at most $\deg(f)$ roots in $\mathbb{K}$.
- (b) If $g \in \mathbb{K}[x]$ is a non-zero polynomial then $f \in \langle g \rangle$ if and only if $r = 0$.

Solution.

(a) **TODO:**

(b) ($\implies$)

Assume $f \in \langle g \rangle$. Then there exists some $q \in \mathbb{K}[x]$ such that $f = qg$. Then,

$$f = qg + r \implies f - qg = r = 0$$

($\impliedby$)

Assume $r = 0$. Then by Theorem 1 we have,

$$f = qg + 0 = qg$$

That is there exists some $q \in \mathbb{K}[x]$ such that $f = qg$ and by definition of an ideal, $f \in \langle g \rangle$.

Problem 2. Implement the algorithm above in Macaulay2, and use your code to compute q and r for the following polynomials.

(a) $f = x^3 + x + 1, \quad g = x + 1.$

(d) $f = x^{100} - 1, \quad g = x^2 - 1.$

(b) $f = x^5 - 1, \quad g = x - 1.$

(e) $f = x^{1000} - 1, \quad g = x - 1.$

(c) $f = x^2 + 1, \quad g = x^3 + x.$

(f) $f = \sum_{i=0}^{500} x^i, \quad g = x^2 + x + 1.$

Solution.

(a) $(x^2 - x + 2, -1)$

(d) $\left(\sum_{k=0}^{49} x^{2k}, 0\right)$

(b) $(x^4 + x^3 + x^2 + 1, 0)$

(e) $\left(\sum_{k=0}^{99} x^k, 0\right)$

(c) $(0, x^2 + 1)$

(f) $\left(\sum_{k=0}^{166} x^{3k}, 0\right)$

Definition 0.2. A monomial ordering on $\mathbb{K}[x_1, \dots, x_n]$ is a relation $<$ on $\mathbb{Z}_{\geq 0}^n$ satisfying the following:

- (i) $<$ is a total ordering on $\mathbb{Z}_{\geq 0}^n$,
- (ii) if $\alpha < \beta$ and $\gamma \in \mathbb{Z}_{\geq 0}^n$ then $\alpha + \gamma < \beta + \gamma$, and
- (iii) $<$ is a well-ordering on $\mathbb{Z}_{\geq 0}^n$.

Recall that a *total ordering* means that of any two elements $\alpha, \beta \in \mathbb{Z}_{\geq 0}^n$ exactly one of the following statements is true: $\alpha < \beta$, $\alpha = \beta$, or $\beta < \alpha$. That $<$ is a well-ordering on $\mathbb{Z}_{\geq 0}^n$ means that every non-empty subset of $\mathbb{Z}_{\geq 0}^n$ has a smallest element with respect to $<$.

Once a monomial ordering has been fixed, every non-zero polynomial acquires a distinguished piece of data: its leading monomial, leading coefficient, and leading term. These are the ingredients that drive the multivariable division process just as degree and leading coefficient drive ordinary polynomial division.

Problem 3. Prove that a monomial ordering $<$ on $\mathbb{Z}_{\geq 0}^n$ is equivalent to a relation $<$ on the set of monomials in $\mathbb{K}[x_1, \dots, x_n]$ satisfying analogous conditions to (i), (ii), and (iii). Explain why this justifies the moniker “monomial ordering”.

Solution.

Problem 4. Prove or disprove that the following are monomial orderings:

- (a) Lexicographic (Lex) Order: Given $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n)$ in $\mathbb{Z}_{\geq 0}^n$, we say $\beta <_{\text{lex}} \alpha$ if and only if the left-most non-zero entry of $\alpha - \beta$ is positive.
- (b) Graded Lex Order: $\beta <_{\text{grlex}} \alpha$ if and only if

$$|\beta| = \sum_{i=1}^n \beta_i < |\alpha| = \sum_{i=1}^n \alpha_i$$

or $|\beta| = |\alpha|$ and $\beta <_{\text{lex}} \alpha$.

- (c) Reverse Lex (RevLex) Order: $\beta <_{\text{revlex}} \alpha$ if and only if the right-most non-zero entry of $\alpha - \beta$ is negative.
- (d) Graded RevLex Order: $\beta <_{\text{grevlex}} \alpha$ if and only if

$$|\beta| = \sum_{i=1}^n \beta_i < |\alpha| = \sum_{i=1}^n \alpha_i$$

or $|\beta| = |\alpha|$ and $\beta <_{\text{revlex}} \alpha$.

- (e) Weight Order: Fix $w = (w_1, \dots, w_n) \in \mathbb{Z}^n$. Define $\beta <_w \alpha$ if and only if

$$\beta \cdot w = \sum_{i=1}^n \beta_i w_i < \alpha \cdot w = \sum_{i=1}^n \alpha_i w_i.$$

Problem 5. Leading Terms. Fix a monomial ordering $<$ on $\mathbb{Z}_{\geq 0}^n$. Let $f = \sum_{\alpha} c_{\alpha} x^{\alpha} \in K[x_1, \dots, x_n]$ be a non-zero polynomial. Define the leading monomial $\text{LM}(f)$, leading coefficient $\text{LC}(f)$, and leading term $\text{LT}(f)$.

(a) Compute $\text{LM}(f)$, $\text{LC}(f)$, and $\text{LT}(f)$ for

$$f = x^3 + 3x^2y + 5xy^3 - 7z^2 \in K[x, y, z]$$

with respect to Lex, Graded Lex, and Graded RevLex (assume $x > y > z$).

(b) Prove that if $f, g \in K[x_1, \dots, x_n]$ are non-zero, then

$$\text{LM}(fg) = \text{LM}(f)\text{LM}(g), \quad \text{LT}(fg) = \text{LT}(f)\text{LT}(g).$$

(c) Explain which property of a monomial ordering is used in part (b).

Problem 6. Let $G = (g_1, \dots, g_s)$ be an ordered list of non-zero polynomials in $K[x_1, \dots, x_n]$.

(a) Write pseudocode for a division algorithm which takes as input f and G and returns polynomials $q_1, \dots, q_s, r$ such that

$$f = q_1 g_1 + \dots + q_s g_s + r,$$

where no monomial in r is divisible by any $\text{LM}(g_1), \dots, \text{LM}(g_s)$.

(b) In the algorithm above, what should happen if the leading monomial of the current polynomial is not divisible by any $\text{LM}(g_i)$?

(c) Prove that your algorithm terminates. What plays the role of “degree” in the proof?

Problem 7. Work in $K[x, y]$ with Lex order and $x > y$. Let $f = xy^2 - x$, $g_1 = xy - 1$, and $g_2 = y^2 - 1$.

(a) Divide f by the ordered list (g_1, g_2) .

(b) Divide f by the ordered list (g_2, g_1) .

(c) Show directly that $f \in \langle g_1, g_2 \rangle$.

(d) What do your computations show about division in several variables? In particular, is the remainder uniquely determined by the ideal $\langle g_1, g_2 \rangle$?

Worksheet 2.1: Zariski Topology

Problem 1. Recall that the spectrum and maximal spectrum of a ring R are the sets:

$$\text{Spec}(R) := \{P \subset R \mid P \text{ is a prime ideal}\}$$

and

$$\text{mSpec}(R) := \{P \subset R \mid P \text{ is a maximal ideal}\}$$

Given any subset $S \subset R$ define

$$\mathbb{V}(S) := \{P \in \text{Spec}(R) \mid S \subset P\}.$$

- (a) Describe $\text{Spec}(\mathbb{K})$ and $\text{mSpec}(\mathbb{K})$ when $\mathbb{K}$ is a field.
- (b) Describe both $\text{Spec}(\mathbb{K}[x]/\langle x^3 \rangle)$ and $\text{mSpec}(\mathbb{K}[x]/\langle x^3 \rangle)$ when $\mathbb{K}$ is a field.
- (c) For an arbitrary ring R , compute $\mathbb{V}(\{0\})$ and $\mathbb{V}(\{1\})$.
- (d) Explain the natural poset structure on $\text{Spec}(R)$. Describe the poset for $\text{Spec}(\mathbb{Z})$.

Solution.

- (a) Let $\mathbb{K}$ be a field. Then $\mathbb{K}$ only has 1 proper ideal: the ideal generated by 0 . Note that this ideal is prime because $\mathbb{K}$ is an integral domain. Moreover it is also maximal because $\mathbb{K}/\langle 0 \rangle = \mathbb{K}$ is a field. So,

$$\text{Spec}(\mathbb{K}) = \text{mSpec}(\mathbb{K}) = \{\langle 0 \rangle\}$$

- (b) Note that the ideal generated by 0 is not prime in this ring because $x \cdot x^2 \in \langle 0 \rangle$ even though neither x nor x^2 are in $\langle x \rangle$. Then we get,

$$\text{Spec}(\mathbb{K}[x]/\langle x^3 \rangle) = \text{mSpec}(\mathbb{K}[x]/\langle x^3 \rangle) = \{\langle x \rangle\}$$

- (c) Let R be an arbitrary ring, then,

$$\mathbb{V}(\{0\}) = \text{Spec}(R)$$

since every ideal contains 0 and,

$$\mathbb{V}(\{1\}) = \emptyset$$

since 1 is not contained in any proper ideal of R .

- (d) The natural poset structure on $\text{Spec}(R)$ is given by inclusion of ideals, that is, for $P_1, P_2 \in \text{Spec}(R)$, we write $P_1 \leq P_2$ iff $P_1 \subseteq P_2$.

The poset for $\text{Spec}(\mathbb{Z})$ contains prime ideals in $\mathbb{Z}$. The zero ideal is contained in every other ideal and every other prime ideal is maximal.

A *topological space* is a set X together with a collection of subsets τ , which satisfies the following conditions: i) $\emptyset \in \tau$ and $X \in \tau$, ii) τ is closed under arbitrary unions, and iii) τ is closed under finite intersections. We call τ a *topology* on X , the elements of τ the open subsets of X , and a subset $Z \subset X$ is closed if and only if $X \setminus Z$ is open (i.e., $X \setminus Z \in \tau$). If $S \subset X$ is a subset, the *subspace topology* on S is the topology where the open sets of S are all sets of the form $S \cap U \subset S$ where $U \subset X$ is open in X . A map $f : X \rightarrow Y$ of topological spaces is *continuous* if and only if $f^{-1}(U) \subset X$ is open for all open subsets $U \subset Y$, i.e., the preimage of an open set is open.

Problem 2. The Zariski Topology. We will now place a topology on the set $\text{Spec}(\mathbf{R})$, called the Zariski topology after Oscar Zariski.

- (a) Prove that if $\mathcal{S}, \mathcal{T} \subset \mathbf{R}$ are arbitrary sets then $\mathbb{V}(\mathcal{S}) \cap \mathbb{V}(\mathcal{T}) = \mathbb{V}(\mathcal{S} \cup \mathcal{T})$. Explain why your argument extends to an *arbitrary* intersection.
- (b) Prove that if $\mathcal{S}, \mathcal{T} \subset \mathbf{R}$ are arbitrary sets then $\mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T}) = \mathbb{V}(\mathcal{S} \cdot \mathcal{T})$ where

$$\mathcal{S} \cdot \mathcal{T} := \{st \mid s \in \mathcal{S}, t \in \mathcal{T}\}.$$

- (c) Prove that the sets of the form $\mathbb{V}(\mathcal{S})$ are the closed sets of a topology on $\text{Spec}(\mathbf{R})$. We call this topology the Zariski topology on $\text{Spec}(\mathbf{R})$. (Hint: Rephrase the definition of a topology in terms of closed sets instead of open sets.)
- (d) Take a break and Google Oscar Zariski and read about his life.

Solution.

- (a) We prove this for arbitrary intersections. Let $\mathcal{S}_i, i \in \Lambda$ where Λ is some indexing set, be arbitrary subsets of $\mathbf{R}$. Then by definition,

$$\bigcap_{i \in \Lambda} \mathbb{V}(\mathcal{S}_i) = \bigcap_{i \in \Lambda} \{[P] \in \text{Spec}(\mathbf{R}), \mathcal{S}_i \subseteq P\} = \left\{ [P] \in \text{Spec}(\mathbf{R}), \bigcup_{i \in \Lambda} \mathcal{S}_i \subseteq P \right\}$$

And

$$\mathbb{V} \left(\bigcup_{i \in \Lambda} \mathcal{S}_i \right) = \left\{ [P] \in \text{Spec}(\mathbf{R}), \bigcup_{i \in \Lambda} \mathcal{S}_i \subseteq P \right\}$$

That is, $\bigcap_{i \in \Lambda} \mathbb{V}(\mathcal{S}_i) = \mathbb{V} \left(\bigcup_{i \in \Lambda} \mathcal{S}_i \right)$. Hence proved.

(b) Let $\mathcal{S}, \mathcal{T}$ be arbitrary subsets of $\mathbf{R}$. Then by definition,

$$\mathbb{V}(\mathcal{S} \cdot \mathcal{T}) = \{[\mathbf{P}] \in \text{Spec}(\mathbf{R}), \mathcal{S} \cdot \mathcal{T} \subseteq \mathbf{P}\}$$

And

$$\mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T}) = \{[\mathbf{P}] \in \text{Spec}(\mathbf{R}), \mathcal{S} \subseteq \mathbf{P}\} \cup \{[\mathbf{P}] \in \text{Spec}(\mathbf{R}), \mathcal{T} \subseteq \mathbf{P}\}$$

Note that $\mathcal{S} \cdot \mathcal{T} \subseteq \mathcal{S}, \mathcal{T}$. Then the containment $\mathbb{V}(\mathcal{S} \cdot \mathcal{T}) \subseteq \mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T})$ is obvious.

To prove that $\mathbb{V}(\mathcal{S} \cdot \mathcal{T}) \supseteq \mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T})$, it suffices to show that if $\mathcal{S} \cdot \mathcal{T} \subseteq \mathbf{P}$ then either $\mathcal{S} \subseteq \mathbf{P}$ or $\mathcal{T} \subseteq \mathbf{P}$.

Assume $\mathcal{S} \cdot \mathcal{T} \subseteq \mathbf{P}$. Let $s \in \mathcal{S}$ and $t \in \mathcal{T}$. Then, since $\mathbf{P}$ is a prime ideal and $st \in \mathbf{P}$ then either $s \in \mathbf{P}$ or $t \in \mathbf{P}$. If both $s, t \in \mathbf{P}$ we are done. Else wlog, assume that $s \notin \mathbf{P}$. But note that $\forall t' \in \mathcal{T}, s \cdot t' \in \mathbf{P}$. Hence, $t' \in \mathbf{P}, \forall t' \in \mathcal{T}$, i.e. $\mathcal{T} \subseteq \mathbf{P}$.

Hence, $\mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T}) = \mathbb{V}(\mathcal{S} \cdot \mathcal{T})$.

(c) To prove that the sets of the form $\mathbb{V}(\mathcal{S})$ are the closed sets of the Zariski topology on $\text{Spec}(\mathbf{R})$ we must show that (i) $\emptyset, \text{Spec}(\mathbf{R})$ are closed sets, (ii) the finite union of closed sets is closed, and (iii) arbitrary intersection of closed sets is closed.

(i) Note that $\emptyset = \{\mathbb{V}(\{1\})\}$ since 1 is not contained in any proper ideal of $\mathbf{R}$, and $\text{Spec}(\mathbf{R}) = \{\mathbb{V}(\{0\})\}$ since 0 is contained in every ideal and thus in every prime ideal. Both $\mathbb{V}(\{1\})$ and $\mathbb{V}(\{0\})$ are closed sets, hence $\emptyset$ and $\text{Spec}(\mathbf{R})$ are closed.

(ii) From part (b) we know that the finite union of two closed sets is closed, and by induction we can extend this to a finite number of closed sets. Hence, the finite union of closed sets is closed.

(iii) From part (a) we know that any arbitrary intersection of closed sets is closed.

Hence, sets of the form $\mathbb{V}(\mathcal{S})$ are the closed sets of the Zariski topology on $\text{Spec}(\mathbf{R})$.

Problem 3. Describe the closed subsets of $\text{mSpec}(\mathbf{R}) \subset \text{Spec}(\mathbf{R})$ in the subspace topology.

Solution.

Closed subsets of $\text{mSpec}(\mathbf{R})$ look like $\mathbb{V}(\mathcal{S}) \cap \text{mSpec}(\mathbf{R})$, for all $\mathbb{V}(\mathcal{S})$ in the Zariski Topology on $\text{Spec}(\mathbf{R})$. That is, the closed subsets of $\text{mSpec}(\mathbf{R})$ are maximal ideals $\mathbf{M}$ such that $\mathcal{S} \subseteq \mathbf{M}$.

Problem 4. Let $\mathbb{K}$ be an algebraically closed field. This exercise will describe the Zariski topology on $\text{Spec}(\mathbb{K}[x])$.

- (a) Describe the points of $\text{Spec}(\mathbb{K}[x])$. (Hint: There are two types.)
- (b) Prove that if $\mathcal{S} \subset \mathbb{K}[x]$ then $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\{f\})$ where f is the greatest common divisor of all elements of $\mathcal{S}$.
- (c) Show that the subspace topology on $\text{mSpec}(\mathbb{K}[x])$ is the finite complement topology. (Recall the *finite complement topology* on a set X is defined by letting the open sets be the empty set and sets of the form $X \setminus T$ for $T \subset X$ a subset of finite cardinality.)

Solution.

- (a) Points of $\text{Spec}(\mathbb{K}[x]) = \{\langle 0 \rangle\} \cup \{\langle x - a \rangle, a \in \mathbb{K}\}$.
- (b) Note that since every element $s \in \mathcal{S}$ is a multiple of the $\text{gcd}(\mathcal{S})$, the containment $\mathbb{V}(\mathcal{S}) \subseteq \mathbb{V}(\{f\})$ is obvious. To show the opposite containment we use **TODO**:
- (c) **TODO**:

Problem 5. Dense Points. If X is a topological space we say that a set $S \subset X$ is *dense* if $U \cap S \neq \emptyset$ for every non-empty open subset $U \subset X$.

- (a) Let $\mathbf{p}$ be a point in a topological space X . Prove that $\{\mathbf{p}\}$ is a dense subset of X if and only if the smallest closed subset which contains $\{\mathbf{p}\}$ is X . We call such points *dense points* of X . (Note: no point in $\mathbb{R}^n$ or $\mathbb{C}^n$ with the Euclidean topology is dense in this sense.)
- (b) Given a prime ideal $\mathbf{P} \subset \mathbf{R}$ we will frequently write $[\mathbf{P}]$ to denote the corresponding point in $\text{Spec}(\mathbf{R})$. Prove that a point $[\mathbf{P}] \in \text{Spec}(\mathbf{R})$ is dense if and only if $\mathbb{V}(\mathbf{P}) = \text{Spec}(\mathbf{R})$.
- (c) Characterize all of the dense points in the following:

- (i) $\text{Spec}(\mathbb{K}[x])$
 - (ii) $\text{Spec}(\mathbb{Z})$
 - (iii) $\text{Spec}(\mathbb{Z}[t])$
 - (iv) $\text{Spec}(\mathbb{K}[x]/\langle x^2 \rangle)$
 - (v) $\text{Spec}(\mathbb{K}[x, y])$
 - (vi) $\text{Spec}(\mathbb{Z}/\langle 6 \rangle)$.
 - (vii) $\text{Spec}(\mathbb{Z}/\langle 4 \rangle)$.
 - (viii) $\text{Spec}(\mathbb{Z}/\langle 12 \rangle)$.
- (d) Prove that if R is a domain then $\text{Spec}(R)$ has a unique dense point.
- (e) Repeat part 5(c) replacing Spec with mSpec in the subspace topology. Can $\text{mSpec}(R)$ ever have a dense point?
- (f) Consider the following two properties (which together are implied by the Hausdorff condition) for a topological space X : i) for every $p \in X$ the set $\{p\}$ is closed and ii) given two distinct points $p, q \in X$ there exist open subsets $U, V \subset X$ such that $p \in U$ and $q \in V$ and $U \cap V = \emptyset$. Prove that if X is a topological space with a dense point, and X is not a single point, then X fails both parts of this definition.

Solution.

1. ($\implies$)

Assume that $\{p\}$ is a dense subset of X .

Then $U \cap \{p\} \neq \emptyset$ for all non-empty open subsets $U \subset X$. That is $\{p\} \not\subseteq X/U$ (since all non-empty proper closed subsets of X are complements of non-empty open subsets of X , $\{p\}$ is not contained in any proper closed subset of X). But $\{p\} \subset X$, hence smallest closed subset which contains $\{p\}$ is X .

($\impliedby$)

Assume that the smallest closed subset which contains $\{p\}$ is X .

Then for all proper closed subsets U' of X , $\{p\} \not\subseteq U'$ that is $X/U' = U \cap \{p\} \neq \emptyset$. Note that $\emptyset \neq U$ for any U' since U' is a proper subset of X . Therefore, for every non-empty open subset $U \subset X$, $\{p\} \cap U \neq \emptyset$, and hence $\{p\}$ is dense in X .

Hence proved that $\{p\}$ is a dense subset of X if and only if the smallest closed subset which contains $\{p\}$ is X .

2. ($\implies$)

Assume $[P] \in \text{Spec}(\mathbf{R})$ is dense. Then the smallest closed set containing $[P]$ is the whole space, i.e. $\text{Spec}(\mathbf{R})$. That is $\mathbb{V}(P) = \text{Spec}(\mathbf{R})$.

($\Leftarrow$)

Assume $\mathbb{V}(P) = \text{Spec}(\mathbf{R})$. Then $P \subseteq T$ for all prime ideals $T \subset \mathbf{R}$. That is, the smallest closed set containing P is the whole space ($\text{Spec}(\mathbf{R})$). Hence, $[P]$ is dense in $\text{Spec}(\mathbf{R})$. '

Hence proved that a point $[P] \in \text{Spec}(\mathbf{R})$ is dense if and only if $\mathbb{V}(P) = \text{Spec}(\mathbf{R})$.

3. **TODO:**

(i)

4. Assume that $\mathbf{R}$ is an integral domain.

Existence: Note that $\langle 0 \rangle \subseteq \mathbf{R}$ is a prime ideal, and also $\langle 0 \rangle \subseteq P$ for all prime ideals P in $\mathbf{R}$. Also recall that the smallest closed set containing $[0]$ is $\mathbb{V}([0])$, and

$$\mathbb{V}(\{[0]\}) = \mathbb{V}(\langle 0 \rangle) = \text{Spec}(\mathbf{R})$$

That is the smallest closed set containing $[0]$ is the whole space, hence $[0]$ is a dense point in $\text{Spec}(\mathbf{R})$.

Uniqueness: Suppose $[P] \in \text{Spec}(\mathbf{R})$ is also a dense point. Then by part(b), we know that $\mathbb{V}(\{[P]\}) = \text{Spec}(\mathbf{R})$. That is $P \subseteq \langle 0 \rangle$, and $\langle 0 \rangle \subseteq P$ trivially. Hence, $[P] = [0]$.

Hence if $\mathbf{R}$ is a domain, $\text{Spec}(\mathbf{R})$ has a unique dense point.

5. **TODO:**

(i)

6. Assume X is a topological space with a dense point $\mathbf{p}$ and X is not a single point.

Then by definition of denseness $\mathbf{p} \in \mathcal{U}$, for all non-empty open subsets $\mathcal{U} \subset X$, i.e. we cannot find two distinct open subsets V, W such that $V \cap W = \emptyset$.

For the sake of contradiction assume $\{\mathbf{p}\}$ is a closed set, then $X \setminus \{\mathbf{p}\}$ is a non-empty open set (since X is not a single point) not containing $\mathbf{p}$ contradicting the definition of denseness. Hence, $\{\mathbf{p}\}$ cannot be closed.

Problem 6. Let $\mathcal{S} \subset \mathbf{R}$ be any subset. Prove that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle)$ where $\langle \mathcal{S} \rangle$ denotes the ideal generated by $\mathcal{S}$.

Solution.

Let I be an ideal containing $\mathcal{S}$, then one can see that $\langle \mathcal{S} \rangle \subseteq I$. And trivially $\mathcal{S} \subset \langle \mathcal{S} \rangle \subseteq I$. Hence,

$$\mathcal{S} \subseteq I \iff \langle \mathcal{S} \rangle \subseteq I$$

And it follows that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle)$.

Problem 7. In this exercise we wish to understand the Zariski topology on $\text{Spec}(\mathbb{C}[x, y])$. Fix a point $\mathbf{p} = (a, b) \in \mathbb{C}^2$.

- (a) Use the *evaluation-at-p* map: $\text{ev} : \mathbb{C}[x, y] \rightarrow \mathbb{C}$ to prove that the set of polynomial functions vanishing at $\mathbf{p}$ is a maximal ideal $\mathfrak{m}_{\mathbf{p}} \subset \mathbb{C}[x, y]$.
- (b) Find generators for the ideal $\mathfrak{m}_{\mathbf{p}}$. Is this ideal principal?
- (c) Prove that the following map is injective:

$$\begin{array}{ccc} \mathbb{C}^2 & \xrightarrow{\eta} & \text{Spec}(\mathbb{C}[x, y]) \\ \mathbf{p} & \longmapsto & \mathfrak{m}_{\mathbf{p}} \end{array}.$$

- (d) Show that the map η from part 3c is not surjective by finding at least two distinct points in $\text{Spec}(\mathbb{C}[x, y])$ not in the image of η .
- (e) Describe the closed subsets of $\text{Spec}(\mathbb{C}[x, y])$. (Hint: They come in four types)

Solution.

(a)

A function $f : X \rightarrow Y$ of topological spaces is *continuous* if and only if $f^{-1}(\mathcal{U}) \subset X$ is open for all open subsets $\mathcal{U} \subset Y$, i.e., the preimage of an open is open. A *homeomorphism* $f : X \rightarrow Y$ is a continuous map of topological spaces with a continuous inverse $f^{-1} : Y \rightarrow X$. A useful fact is that a continuous map $f : X \rightarrow Y$ is a homeomorphism if and only if it is bijective and $f(\mathcal{U}) \subset Y$ is open for every open subsets $\mathcal{U} \subset X$.

Problem 8. Ring Maps Induce Continuous Maps. Let $\phi : R \rightarrow S$ be a ring homomorphism. The goal of this exercise is to show ϕ induces a map $\phi^\# : \text{Spec}(S) \rightarrow \text{Spec}(R)$, which is continuous in the Zariski topology.

- (a) Prove that if $[P] \in \text{Spec}(S)$, then $[\phi^{-1}(P)] \in \text{Spec}(R)$.
- (b) Prove that the map below is continuous in the Zariski topology:

$$\begin{array}{ccc} \text{Spec}(S) & \xrightarrow{\phi^\#} & \text{Spec}(R) \\ [P] & \longmapsto & [\phi^{-1}(P)] \end{array}$$

- (c) For the ring map $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ sending each integer to its residue class modulo 2, describe the induced map on spectra explicitly.
- (d) For the ring map $\mathbb{Z} \hookrightarrow \mathbb{Q}$, describe the induced map on spectra explicitly.
- (e) If $\phi : R \hookrightarrow S$ is an injective ring map, give a simple description of the induced map on spectra.

Solution.

- (a) **Claim:** Let $\phi : R \rightarrow S$ be a ring homomorphism, and $P \subseteq S$ be a prime ideal, then $\phi^{-1}(P)$ is a prime ideal.

Proof:

A *category* is a collection of objects together with, for each pair of objects, a set of morphisms between them, equipped with an associative composition law and an identity morphism for each object. A *functor* between categories **C** and **D** is a mapping that assigns to each object of **C** an object of **D** and to each morphism in **C** a morphism in **D**, preserving identity morphisms and composition. A functor is *contravariant* if it reverses the order of composition and *covariant* if it preserves the order of composition.

Problem 9. Spec as a Functor. In this problem we package together much of this worksheet to define the Spec functor. Consider the map:

$$\begin{array}{ccc} \mathbf{comRing} & \longrightarrow & \mathbf{Top} \\ R & \longmapsto & \text{Spec}(R) \end{array}$$

from the category of commutative rings with unit with ring homomorphisms to the category of topological spaces with continuous maps.

- (a) Explain using Question ?? how this mapping defines a contravariant functor from **comRing** to **Top**. To check functoriality, verify

$$(\text{Id}_R)^\# = \text{Id}_{\text{Spec}(R)} \quad \text{and} \quad (\psi \circ \phi)^\# = \phi^\# \circ \psi^\#$$

for ring maps $\phi : R \rightarrow S$ and $\psi : S \rightarrow T$. We call this the Spec functor.

- (b) Do isomorphic rings have Spectra that are homeomorphic topological spaces? Explain.
- (c) If two rings have homeomorphic spectra, are they necessarily isomorphic rings? Explain.

Problem 10. Find a homeomorphism between $\text{Spec}(\mathbb{C}[x])$ and $\text{Spec}(\mathbb{C}[x, y]/\langle x - y \rangle)$. Construct the corresponding ring map explicitly in *Macaulay2* and compute its kernel.

Problem 11. Closed Embeddings I. Let $I \subset R$ be an ideal and $\pi : R \rightarrow R/I$ the natural quotient map. Consider the induced map on spectra

$$\text{Spec}(R/I) \xrightarrow{\pi^\#} \text{Spec}(R)$$

In this exercise we will show this map is a closed embedding. A *closed embedding* is a map of topological spaces whose image is closed and which is a homeomorphism onto its image.

- (a) Using the correspondence between ideals in R and R/I give an explicit description of $\pi^\#$.
- (b) Prove that the image of $\pi^\#$ is equal to $\mathbb{V}(I)$.
- (c) Explain why the ideal correspondence implies that $\pi^\#$ is injective.
- (d) Prove that $\pi^\#$ is a closed embedding onto $\mathbb{V}(I)$, in the subspace topology, by showing that if $J \subset R/I$ is an ideal then

$$\pi^\#(\mathbb{V}(J)) = \mathbb{V}(\pi^{-1}(J)).$$

Worksheet 2.2: Algebraic Sets

Theorem 0.3 (Hilbert’s Nullstellensatz I). *If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1, \dots, x_n]$ is of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$ for some point $(a_1, \dots, a_n) \in \mathbb{K}^n$.*

Note this is just the first of a series of theorems that will go by Hilbert’s Nullstellensatz. Recall that a $\mathbb{K}$ -algebra is a ring R together with a specified ring homomorphism $\mathbb{K} \rightarrow R$. A morphism of $\mathbb{K}$ -algebras $\varphi : R \rightarrow S$ is a ring homomorphism that preserves scalars, i.e. $\varphi(c \cdot 1_R) = c \cdot 1_S$ for every $c \in \mathbb{K}$. In particular, $\mathbb{K}[x_1, \dots, x_n]$ is a $\mathbb{K}$ -algebra, every evaluation map $\text{ev}_p : \mathbb{K}[x_1, \dots, x_n] \rightarrow \mathbb{K}$ is a $\mathbb{K}$ -algebra homomorphism, and if $\mathfrak{m}$ is a maximal ideal then $R/\mathfrak{m}$ is naturally a field equipped with a $\mathbb{K}$ -algebra structure. When we say below that a residue field is “isomorphic to $\mathbb{K}$,” we mean *as a $\mathbb{K}$ -algebra*.

Problem 1. Evaluation Maps. For a point $p = (a_1, \dots, a_n) \in \mathbb{K}^n$ the *evaluation-at- p* map is given by

$$\begin{array}{ccc} \mathbb{K}[x_1, \dots, x_n] & \xrightarrow{\text{ev}_p} & \mathbb{K} \\ f & \longmapsto & f(p) \end{array}.$$

- (a) Prove that ev_p is a ring homomorphism and find generators for $\mathfrak{m}_p := \ker(\text{ev}_p)$.
- (b) Prove that there is a bijection, induced by these evaluation maps, between $\mathbb{K}^n$ and the subset of $\text{Spec}(\mathbb{K}[x_1, \dots, x_n])$ consisting of maximal ideals whose residue field is isomorphic to $\mathbb{K}$ as a $\mathbb{K}$ -algebra.
- (c) Find a maximal ideal of $\mathbb{R}[x, y]$ that is not of the form $\mathfrak{m}_p$ for any $p \in \mathbb{R}^2$.
- (d) Restate Hilbert’s Nullstellensatz I (above) in terms of $\mathfrak{m}_p$.

Solution.

- (a) To show that ev_p is a ring homomorphism, we need to show that for all $f_1, f_2 \in \mathbb{K}[x_1, \dots, x_n]$ the three conditions hold,

Additivity:

$$\text{ev}_p(f_1 + f_2) = (f_1 + f_2)(p) = f_1(p) + f_2(p) = \text{ev}_p(f_1) + \text{ev}_p(f_2)$$

Multiplicativity:

$$\text{ev}_p(f_1 \cdot f_2) = (f_1 \cdot f_2)(p) = f_1(p) \cdot f_2(p) = \text{ev}_p(f_1) \cdot \text{ev}_p(f_2)$$

And $\text{ev}_p(1) = 1$ since 1 is a constant function.

(b) **TODO:**

(c) **TODO:**

(d) **TODO:**

Problem 2. Algebraic Sets. Let $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$ be a subset. Define the corresponding *algebraic set* in $\mathbb{K}^n$ to be:

$$\mathbb{V}(\mathcal{S}) := \{ \mathbf{p} \in \mathbb{K}^n \mid f(\mathbf{p}) = 0 \text{ for all } f \in \mathcal{S} \}.$$

(a) Prove that $\mathbb{V}$ is order-reverse, i.e., if $\mathcal{S} \subset \mathcal{T}$ then $\mathbb{V}(\mathcal{T}) \subset \mathbb{V}(\mathcal{S})$.

(b) Prove that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle) = \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle})$.

(c) Describe each of the following algebraic sets, and when possible draw a picture.

(i) $\mathcal{S} := \{x^2 - y\} \subset \mathbb{R}[x, y]$.

(ii) $\mathcal{S} := \{x^2 - xy\} \subset \mathbb{R}[x, y]$.

(iii) $\mathcal{S} := \{x^6 - 3x^5y + 3x^4y^2 - x^3y^3\} \subset \mathbb{R}[x, y]$.

(iv) $\mathcal{S} := \{x^4 - 2x^3y + x^2y^2\} \subset \mathbb{R}[x, y]$.

(v) $\mathcal{S} := \{x^2 + y^2 - 1\} \subset \mathbb{R}[x, y]$.

(vi) $\mathcal{S} := \{n(x^2 + y^2 - z^2) \mid n \in \mathbb{Z}_{\geq 0}\} \subset \mathbb{R}[x, y, z]$.

(vii) $\mathcal{S} := \{x^2 + 1\} \subset \mathbb{R}[x]$.

(viii) $\mathcal{S} := \{x^2 + 1\} \subset \mathbb{C}[x]$.

(ix) $\mathcal{S}$ is a collection of any linear polynomials in $\mathbb{K}[x_1, \dots, x_n]$.

(d) Explain why $\mathrm{SL}_n(\mathbb{K})$ has the structure of an algebraic set.

(e) Explain why the set of $m \times n$ matrices of rank $< t$ is an algebraic set contained in $\mathbb{K}^{m \times n}$.

Solution.

(a) Assume $\mathcal{S} \subset \mathcal{T}$. Let $\mathbf{p} \in \mathbb{V}(\mathcal{T})$.

Then, for all $f \in \mathcal{T}$, $f(\mathbf{p}) = 0$. But since $\mathcal{S} \subset \mathcal{T}$ this implies $g(\mathbf{p}) = 0, \forall g \in \mathcal{S}$. Hence, $\mathbf{p} \in \mathbb{V}(\mathcal{S})$. Since $\mathbf{p}$ was an arbitrary point in $\mathbb{V}(\mathcal{T})$ we conclude that $\mathbb{V}$ is order reversing.

(b) By definition,

$$\mathcal{S} \subseteq \langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}.$$

And from part (a) we have,

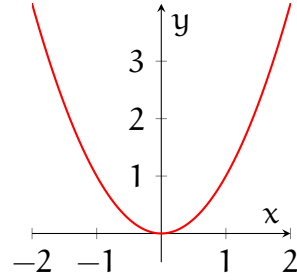
$$\mathbb{V}(\mathcal{S}) \supseteq \mathbb{V}(\langle \mathcal{S} \rangle) \supseteq \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right).$$

To show $\mathbb{V}(\langle \mathcal{S} \rangle) \subseteq \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right)$ we let $\mathbf{p} \in \mathbb{V}(\langle \mathcal{S} \rangle)$. That is for all $f \in \langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}$, $f(\mathbf{p}) = 0$, i.e. $\mathbf{p} \in \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right) \implies \mathbb{V}(\langle \mathcal{S} \rangle) \subseteq \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right)$.

To show that $\mathbb{V}(\mathcal{S}) \subseteq \mathbb{V}(\langle \mathcal{S} \rangle)$ we let $\mathbf{p} \in \mathbb{V}(\mathcal{S})$. That is for all $f \in \mathcal{S}$, $f(\mathbf{p}) = 0$ but since $\langle \mathcal{S} \rangle$ is the set of linear combinations of functions that evaluate to zero at $\mathbf{p}$, they also evaluate to zero at $\mathbf{p}$ and $\mathbf{p} \in \mathbb{V}(\langle \mathcal{S} \rangle)$.

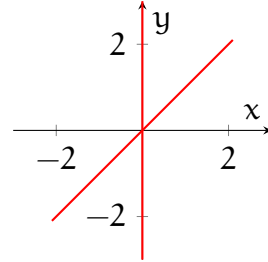
- (c) (i) $\mathcal{S} := \{x^2 - y\} \subset \mathbb{R}[x, y]$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y\}$$



- (ii) $\mathcal{S} := \{x^2 - xy\} \subset \mathbb{R}[x, y]$.

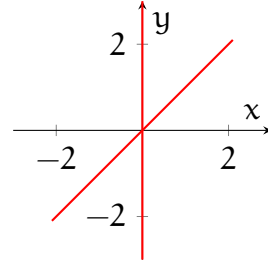
$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0, x = y\}$$



- (iii) $\mathcal{S} := \{x^6 - 3x^5y + 3x^4y^2 - x^3y^3\} \subset \mathbb{R}[x, y]$.

Note that $x^6 - 3x^5y + 3x^4y^2 - x^3y^3 = x^3(x - y)^3$.

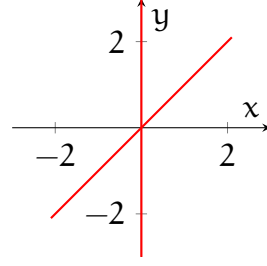
$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0, x = y\}$$



- (iv) $\mathcal{S} := \{x^4 - 2x^3y + x^2y^2\} \subset \mathbb{R}[x, y]$.

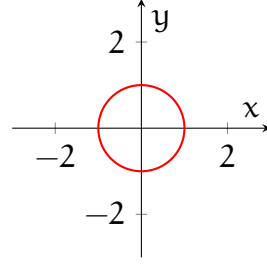
Note that $x^4 - 2x^3y + x^2y^2 = x^2(x - y)^2$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0, x = y\}$$



(v) $\mathcal{S} := \{x^2 + y^2 - 1\} \subset \mathbb{R}[x, y]$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$



(vi) $\mathcal{S} := \{n(x^2 + y^2 - z^2) \mid n \in \mathbb{Z}_{\geq 0}\} \subset \mathbb{R}[x, y, z]$.

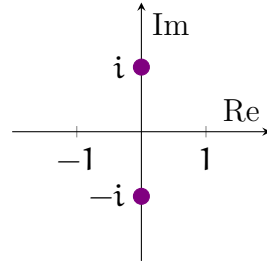
$$\mathbb{V}(\mathcal{S}) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = z^2\}$$

(vii) $\mathcal{S} := \{x^2 + 1\} \subset \mathbb{R}[x]$.

$$\mathbb{V}(\mathcal{S}) = \emptyset \text{ since this equation has no solutions over } \mathbb{R}[x]$$

(viii) $\mathcal{S} := \{x^2 + 1\} \subset \mathbb{C}[x]$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y\}$$



(ix) $\mathcal{S}$ is a collection of any linear polynomials in $\mathbb{K}[x_1, \dots, x_n]$.

$$\mathbb{V}(\mathcal{S}) = \ker(A), \text{ where } A \text{ is a } m \times n \text{ matrix where the } i\text{-th row is } (a_{i_1}, \dots, a_{i_n})$$

(d) $\text{SL}_n(\mathbb{K})$ is the set of $n \times n$ matrices with determinant 1, and the determinant is a polynomial function on $\mathbb{K}^{n \times n}$. Therefore we can define the polynomial

$$f = \det(A) - 1 \in \mathbb{K}[x_1, \dots, x_{n \times n}]$$

where A is a $n \times n$ matrix, and $\text{SL}_n(\mathbb{K}) = \mathbb{V}(\{f\})$.

(e) **TODO:**

Problem 3. Defining Ideal of an Algebraic Set. Let $V \subset \mathbb{K}^n$ be an algebraic set. The *defining ideal* of V is the ideal in $\mathbb{K}[x_1, \dots, x_n]$ given by

$$\mathbb{I}(V) := \{f \in \mathbb{K}[x_1, \dots, x_n] \mid f(\mathbf{p}) = 0 \text{ for all } \mathbf{p} \in V\}.$$

- (a) Prove that $\mathbb{I}(V)$ is in fact an ideal.
- (b) Prove that $\mathbb{I}(V)$ is radical.
- (c) If $V = \mathbb{V}(\mathcal{S})$ for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, show that $\langle \mathcal{S} \rangle \subset \sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$.
- (d) Let $\mathbb{K} = \mathbb{R}$. Give an example to show that the inclusion $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ can be proper.
- (e) If $V = \mathbb{V}(\mathcal{S})$ for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, show that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\mathbb{I}(V))$.

Solution.

- (a) Let $g \in \mathbb{K}[x_1, \dots, x_n]$ and $f_1, f_2 \in \mathbb{I}(V)$. That is, $f_1(\mathbf{p}) = f_2(\mathbf{p}) = 0$.

Subgroup Under Addition: Note that,

$$(f_1 + f_2)(\mathbf{p}) = f_1(\mathbf{p}) + f_2(\mathbf{p}) = 0$$

The zero polynomial $0(\mathbf{p}) = 0$ acts as the identity, and

$$f_1(\mathbf{p}) + (-f_1)(\mathbf{p}) = 0(\mathbf{p}) = 0$$

gives us the additive inverse.

Closed Under Multiplication: Notice,

$$(g \cdot f)(\mathbf{p}) = g(\mathbf{p}) \cdot f_1(\mathbf{p}) = g(\mathbf{p}) \cdot 0 = 0$$

Thus $g \cdot f \in \mathbb{I}(V)$ and $\mathbb{I}(V)$ is an ideal.

- (b) We want to show that $\mathbb{I}(V) = \sqrt{\mathbb{I}(V)}$.

Note that the containment $\mathbb{I}(V) \subseteq \sqrt{\mathbb{I}(V)}$ is obvious. To show the opposite containment take $f \in \sqrt{\mathbb{I}(V)}$. Then for some n we have,

$$f^n(\mathbf{p}) = (f(\mathbf{p}))^n = 0$$

But since $\mathbb{K}$ is a domain, this implies that $f(\mathbf{p}) = 0$. Hence $f \in \mathbb{I}(V)$, $\mathbb{I}(V) \supseteq \sqrt{\mathbb{I}(V)}$, and $\mathbb{I}(V) = \sqrt{\mathbb{I}(V)}$.

- (c) Let $V = V(\mathcal{S})$. We know that $\langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}$. So showing that $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ suffices. Note that $\mathbb{I}(V) = \mathbb{I}(V(\mathcal{S}))$ which is the set of all functions that evaluate to 0 at $\mathbf{p}$, for all $\mathbf{p} \in V(\mathcal{S})$. And $V(\mathcal{S})$ is the set of all points $\mathbf{p}$ such that all functions $f \in \mathcal{S}$ evaluate to 0 at $\mathbf{p}$. Let $f \in \sqrt{\langle \mathcal{S} \rangle}$. Then by definition for some n , $f^n \in \langle \mathcal{S} \rangle$. We can then write,

$$f^n = \sum_i h_i(\mathbf{p}) g_i(\mathbf{p}), \quad h_i \in \mathbb{K}[x_1, \dots, x_n], g_i \in \mathcal{S}$$

But then,

$$f^n(\mathbf{p}) = \sum_i h_i(\mathbf{p}) g_i(\mathbf{p}) = \sum_i h_i(\mathbf{p}) \cdot 0 = 0$$

Moreover since $\mathbb{K}$ is a domain, $f^n(\mathbf{p}) = 0 \implies f(\mathbf{p}) = 0 \implies f$ is a function that evaluates to 0 at $\mathbf{p}$, hence $f \in \mathbb{I}(V)$.

- (d) Consider the set $\mathcal{S} = \{x^2 + 1\}$. We have,

$$\sqrt{\langle \mathcal{S} \rangle} = \sqrt{\langle x^2 + 1 \rangle}$$

And, because the polynomial has no solutions over $\mathbb{R}$, we have $V = V(\mathcal{S}) = \emptyset$. But $\mathbb{I}(V(\mathcal{S})) = \mathbb{R}[x]$. Also note that since $\langle \mathcal{S} \rangle$ is a prime ideal ($x^2 + 1$ is irreducible), we have $\langle x^2 + 1 \rangle = \sqrt{\langle x^2 + 1 \rangle} \subset \mathbb{R}[x]$. Hence, the containment $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ can be proper.

- (e) Let $V = V(\mathcal{S})$.

$$V(\mathcal{S}) \subseteq V(\mathbb{I}(V(\mathcal{S})))$$

This containment is obvious by construction. $\mathbb{I}(V(\mathcal{S}))$ is the set of all functions that vanish at all points in $V(\mathcal{S})$. And $V(\mathbb{I}(V(\mathcal{S})))$ is the set of all points that all functions in $\mathbb{I}(V(\mathcal{S}))$ vanish at. So, $V(\mathcal{S}) \subseteq V(\mathbb{I}(V(\mathcal{S})))$.

$$V(\mathcal{S}) \supseteq V(\mathbb{I}(V(\mathcal{S})))$$

Note that by definition $\mathcal{S} \subseteq \mathbb{I}(V(\mathcal{S}))$. Then by 2(a) $V(\mathbb{I}(V(\mathcal{S}))) \subseteq V(\mathcal{S})$.

Hence, $V(\mathcal{S}) = V(\mathbb{I}(V(\mathcal{S})))$.

So far you have built two order-reversing operations:

$$\mathcal{S} \longmapsto \mathbb{V}(\mathcal{S}) \quad \text{and} \quad V \longmapsto \mathbb{I}(V).$$

You have shown that passing from equations to a zero set only remembers the radical of the ideal they generate, and Question 4d shows that over a non-algebraically closed field this can lose information. The next theorem says that over an algebraically closed field this is the only ambiguity.

Theorem 0.4 (Hilbert's Nullstellensatz II). *If $\mathbb{K}$ is algebraically closed and $V = \mathbb{V}(\mathcal{S}) \subset \mathbb{K}^n$ is an algebraic set, for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, then $\mathbb{I}(V) = \sqrt{\langle \mathcal{S} \rangle}$.*

At this point the basic affine dictionary is taking shape; Question (4) asks you to make the correspondence between algebraic sets and radical ideals completely explicit. After that we turn to coordinate rings, which package an algebraic set V into algebra and will let us recover from $\mathbb{K}[V]$ its points, its topology, and ultimately its morphisms.

Problem 4. Let $\mathbb{K}$ be algebraically closed. Use Hilbert's Nullstellensatz II (above) to give an explicit inclusion-reversing bijection between algebraic sets in $\mathbb{K}^n$ and radical ideals.

Solution.

Let $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$.

$$\mathbb{V}(\mathcal{S}) \xrightarrow{\mathbb{I}} \sqrt{\langle \mathcal{S} \rangle}$$

$$\mathbb{V}(\mathcal{S}) \xleftarrow{\mathbb{V}} \sqrt{\langle \mathcal{S} \rangle}$$

TODO: Diagram.

Problem 5. Coordinate Rings. Let $V \subset \mathbb{K}^n$ be an algebraic set and $\mathcal{R} = \text{Hom}_{\text{Set}}(V, \mathbb{K})$ be the ring of all (set) functions from V to $\mathbb{K}$. In this question we study the restriction map from $\mathbb{K}[x_1, \dots, x_n]$ to $\mathcal{R}$ given by:

$$\begin{array}{ccc} \mathbb{K}[x_1, \dots, x_n] & \xrightarrow{\rho} & \mathcal{R} \\ f & \longmapsto & f|_V \end{array}.$$

Here we are viewing $f \in \mathbb{K}[x_1, \dots, x_n]$ as defining a function $\mathbb{K}^n \rightarrow \mathbb{K}$ given by $\mathbf{p} \mapsto f(\mathbf{p})$. We call the image of ρ the *coordinate ring* of V and denote it $\mathbb{K}[V]$.

- (a) Prove that ρ is in fact a ring homomorphism.
- (b) Verify that each of the following functions lies in the coordinate ring of the given algebraic set.
 - (i) Let $V = \mathbb{V}(x^2 + y^2 - 1) \subset \mathbb{R}^2$ and consider the function $V \rightarrow \mathbb{R}$ given by $(x, y) \mapsto e^{x^2+y^2}$.
 - (ii) Let $V = \mathbb{V}(xy - 1) \subset \mathbb{K}^2$ and consider the function $V \rightarrow \mathbb{K}$ given by $(x, y) \mapsto \frac{1}{x}$.
- (c) Identify the kernel of ρ and use it to give an explicit presentation for $\mathbb{K}[V]$ as a $\mathbb{K}$ -algebra.
- (d) The i -th coordinate of $\mathbb{K}^n$ is the linear functional $e_i^\vee : \mathbb{K}^n \rightarrow \mathbb{K}$ sending $(a_1, \dots, a_n) \mapsto a_i$. Explain why $e_i^\vee \in \mathbb{K}[\mathbb{K}^n]$, and under the presentation given in part (c), which polynomial corresponds to $e_i^\vee$.
- (e) Explain why $\mathbb{K}[V]$ is called the coordinate ring of V . Hint: Think about the previous part.
- (f) Describe the coordinate ring for the following algebraic sets:
 - (i) $\mathbb{V}(y - x^2) \subset \mathbb{K}^2$.
 - (ii) $\mathbb{V}(y^2, xy, x^{17}) \subset \mathbb{K}^2$.
 - (iii) $\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) \subset \mathbb{K}^2$.
 - (iv) The set of 3×3 matrices over $\mathbb{F}_5$ of determinant 1 and trace 0. (Hint/Warning: There will likely be 9 more polynomials in the defining ideal than you expect, coming from the fact that finite fields have a special automorphism. This highlights another way we might get a counterexample for Question 4d.)

Solution.

- (a) Let $f_1, f_2 \in \mathbb{K}[x_1, \dots, x_n]$. Note that addition and multiplication is defined pointwise in $\mathbb{R}$. To see if ρ is a ring homomorphism, we check if the three properties hold:

Addition:

$$\rho(f_1) + \rho(f_2)(p) = f_1|_V(p) + f_2|_V(p) = (f_1 + f_2)|_V(p) = \rho(f_1 + f_2)(p)$$

Multiplication:

$$\rho(f_1) \cdot \rho(f_2)(p) = f_1|_V(p) \cdot f_2|_V(p) = (f_1 \cdot f_2)|_V(p) = \rho(f_1 \cdot f_2)(p)$$

Unital:

$$\rho(1)(p) = 1|_V(p) = 1$$

- (b) (i) Let $f : (x, y) \mapsto e^{x^2+y^2}$. Note that on V , $x^2 + y^2 = 1$ then, $f|_V = e^{x^2+y^2} = e^1$. Which behaves like the constant polynomial function $g(x, y) = e$ and hence lies in the ring of coordinates of V .
- (ii) Let $f : (x, y) \mapsto \frac{1}{x}$. Note that on V , $\frac{1}{x} = y$, and $x \neq 0$ because $\mathbb{K}$ is a domain. Then, $f|_V = \frac{1}{x} = y$. Which behaves like the polynomial function $g(x, y) = y$ and hence lies in the ring of coordinates of V .
- (c) The kernel of ρ is $\mathbb{I}(V)$.

$$\mathbb{K}[V] \cong \mathbb{K}[x_1, \dots, x_n]/\mathbb{I}(V)$$

- (d) The coordinate ring of $\mathbb{K}^n$, $\mathbb{K}[\mathbb{K}^n]$ is the ring of all set functions $\mathbb{K}^n \rightarrow \mathbb{K}$.

$$e_i^\vee(x_1, \dots, x_n) = x_i$$

which corresponds to the polynomial function $g(x_1, \dots, x_n) = x_i$. Hence, $e_i^\vee \in \mathbb{K}[\mathbb{K}^n]$. Under the presentation given in part (c) we have,

$$\mathbb{K}[\mathbb{K}^n] \cong \mathbb{K}[x_1, \dots, x_n]/\mathbb{I}(\mathbb{K}^n) = \mathbb{K}[x_1, \dots, x_n]/\{0\} = \mathbb{K}[x_1, \dots, x_n]$$

So, $e_i^\vee$ corresponds to the coset $[x_i] = x_i$.

- (e) $\mathbb{K}[V]$ is the coordinate ring of V because every polynomial is a combination of coordinate functions restricted to V . And $\mathbb{K}[V]$ is generated by these functions on V .

$$\mathbb{K}[V] = \mathbb{K}[e_1^\vee|_V, \dots, e_n^\vee|_V]$$

- (f) (i) Let $V = \mathbb{V}(y - x^2)$.

$$\mathbb{K}[V] \cong \mathbb{K}[x, y]/\mathbb{I}(y - x^2) = \mathbb{K}[x, x^2] = \mathbb{K}[x]$$

(ii) Let $V = \mathbb{V}(y^2, xy, x^{17})$.

Note that $I(V) = \sqrt{\langle y^2, xy, x^{17} \rangle}$.

Since $\mathbb{K}$ is a field, if $(a, b) \in V$ then

$$a^{17} = 0 \implies a = 0$$

and

$$b^2 = 0 \implies b = 0$$

Then, $V = \{(0, 0)\}$. And $I(V) = I(\{(0, 0)\}) = \langle x, y \rangle$.

$$\mathbb{K}[V] \cong \mathbb{K}[x, y]/I(V) = \mathbb{K}[x, y]/\langle x, y \rangle = \mathbb{K}$$

(iii) Let $V = \mathbb{V}(x^6 - 3x^5y + 3x^4y^2 + x^3y^3) = \mathbb{V}(x^3(x - y)^3)$.

Then if $(a, b) \in V$, either $a^3 = 0 \implies a = 0$ or $(x - y)^3 = 0 \implies x = y$.

Then $V = \{(0, a), (a, a) \mid a \in \mathbb{K}\} = \mathbb{V}(x) \cup \mathbb{V}(x - y)$.

$$I(V) = I(\mathbb{V}(x) \cup \mathbb{V}(x - y)) = I(\mathbb{V}(x)) \cap I(\mathbb{V}(x - y)) = \langle x \rangle \cap \langle x - y \rangle = \langle x(x - y) \rangle$$

And

$$\mathbb{K}[V] \cong \mathbb{K}[x, y]/\langle x(x - y) \rangle$$

(iv) **TODO:**

$$A \in \mathrm{SL}_3(\mathbb{F}_5)$$

$$\mathbb{V}(\mathrm{tr}(A))$$

You should now think of $\mathbb{K}[V]$ as the algebraic avatar of V : its elements are precisely the polynomial functions on V , and the coordinate functions on $\mathbb{K}^n$ descend to generators of $\mathbb{K}[V]$. The next step is to recognize points of V purely algebraically, either as $\mathbb{K}$ -algebra maps $\mathbb{K}[V] \rightarrow \mathbb{K}$ or as maximal ideals whose residue fields are isomorphic to $\mathbb{K}$ as $\mathbb{K}$ -algebras.

Problem 6. Let R be a finitely generated $\mathbb{K}$ -algebra. Fix a presentation $R \cong \mathbb{K}[x_1, \dots, x_n]/I$. Describe a natural bijection between the following three sets:

- (i) points in $V(I) \subset \mathbb{K}^n$,
- (ii) maximal ideals in R with residue field isomorphic to $\mathbb{K}$ as a $\mathbb{K}$ -algebra, and
- (iii) elements of $\text{Hom}_{\mathbb{K}\text{-alg}}(R, \mathbb{K})$, the set of morphisms from R to $\mathbb{K}$ in the category of $\mathbb{K}$ -algebras.

Solution.

Let $\mathbb{K}$ be algebraically closed.

Claim: Let $\pi: \mathbb{K}[x_1, \dots, x_n] \rightarrow \mathbb{K}[x_1, \dots, x_n]/I = R$, and let $J \subset R$ be maximal. We claim that $\pi^{-1}(J)$ is maximal in $\mathbb{K}[x_1, \dots, x_n]$.

Proof. We know that $\pi^{-1}(J)$ is an ideal. Let M be a maximal ideal containing $\pi^{-1}(J)$. Since π is surjective, we know that $\pi(M)$ is an ideal (worksheet 1.1, Q4.4), and $J \subseteq \pi(M)$. Then, since J is maximal, either $\pi(M) = J$ or $\pi(M)$ is the whole ring. $\pi(M) \neq R$ since ring homomorphisms are unital and $1 \notin M$. Hence, $M = \pi^{-1}(J)$ or $\pi^{-1}(J)$ is maximal. $\square$

(i) $\iff$ (ii)

Recall that $V(I)$ contains points $\mathbf{p} = (a_1, \dots, a_n)$ such that $f(\mathbf{p}) = 0$ for all $f \in I$. Also,

$$\text{ev}_{\mathbf{p}}: \mathbb{K}^n \rightarrow \mathbb{K}$$

By Hilbert's Nullstellensatz I, we know that maximal ideals of $\mathbb{K}[x_1, \dots, x_n]$ are of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$, for some point $(a_1, \dots, a_n)$.

Problem 7. Zariski Topology on Algebraic Sets. Let $V \subseteq \mathbb{K}^n$ be an algebraic set. Define the *Zariski topology* on V by declaring the closed sets to be the subsets of V of the form $V(I)$ for some ideal $I \subset \mathbb{K}[x_1, \dots, x_n]$. In other words, the closed sets are precisely the *algebraic subsets* of V .

- (a) Verify that this definition indeed determines a topology on V .

- (b) For $f \in \mathbb{K}[V]$, define

$$D(f) = \{p \in V \mid f(p) \neq 0\}.$$

Show that $\{D(f)\}$ form a basis for the Zariski topology on V .

- (c) For each point $p \in V$, let

$$\mathfrak{m}_p = \{f \in \mathbb{K}[V] \mid f(p) = 0\}$$

be the corresponding maximal ideal. Show that the map

$$\begin{array}{ccc} V & \longrightarrow & \text{Spec}(\mathbb{K}[V]) \\ p & \longmapsto & \mathfrak{m}_p \end{array}$$

is a homeomorphism onto its image, which consists of those maximal ideals whose residue field is isomorphic to $\mathbb{K}$ as a $\mathbb{K}$ -algebra.

- (d) If $\mathbb{K}$ is algebraically closed how can the preceding part be rephrased in terms of $\text{mSpec}(\mathbb{K}[V])$?
- (e) Is the Zariski topology on V Hausdorff in general? If not, when is it?

Solution.

- (a) $V = \mathbb{V}(I)$ for some $I \subset \mathbb{K}[x_1, \dots, x_n]$. The closed sets of V are of the form $\mathbb{V}(J)$, where $I \subset J \subset \mathbb{K}[x_1, \dots, x_n]$. We need to check the following:

- (i) $\emptyset$ and V are closed.

Note $\mathbb{V}(I) = V$, and $\mathbb{V}(\mathbb{K}[x_1, \dots, x_n]) = \emptyset$ is closed.

- (ii) Finite union of closed sets is closed.

Let $\mathbb{V}(J), \mathbb{V}(K) \subseteq V$ be subsets. Then, $\mathbb{V}(I) \cup \mathbb{V}(J)$ is the set of points in $\mathbb{K}^n$ such that for all $p \in \mathbb{V}(I) \cup \mathbb{V}(J)$ either $f(p) = 0$ or $g(p) = 0$.

- (iii) Arbitrary intersection of closed sets is closed.

By now you have recovered from $\mathbb{K}[V]$ not just the points of V , but also its Zariski topology. The next goal is to understand maps in the same way: a morphism of algebraic sets should be encoded contravariantly by a $\mathbb{K}$ -algebra homomorphism of coordinate rings.

Problem 8. Morphisms of Algebraic Sets. A *polynomial map* from $\mathbb{K}^n$ to $\mathbb{K}^m$ is a function of the form:

$$\Phi : \mathbb{K}^n \longrightarrow \mathbb{K}^m \quad \mathbf{p} \mapsto (f_1(\mathbf{p}), \dots, f_m(\mathbf{p})).$$

where $f_1, \dots, f_m \in \mathbb{K}[x_1, \dots, x_n]$. A *morphism of algebraic sets* $V \rightarrow W$ is defined to be the restriction of such a polynomial map Φ with $\Phi(V) \subset W$. Fix a polynomial map Φ as above.

- (a) Show that there is an induced $\mathbb{K}$ -algebra homomorphism

$$\phi : \mathbb{K}[\mathbb{K}^m] \longrightarrow \mathbb{K}[\mathbb{K}^n] \quad \text{given by} \quad g \mapsto g \circ \Phi.$$

Describe ϕ explicitly in terms of the coordinate function generators of $\mathbb{K}[\mathbb{K}^m]$.

- (b) Let $V \subset \mathbb{K}^n$ and $W \subset \mathbb{K}^m$ be algebraic sets. Show that $\Phi(V) \subset W$ if and only if $\phi(\mathbb{I}(W)) \subset \mathbb{I}(V)$.
- (c) Deduce that Φ restricts to a morphism of algebraic sets $\Phi|_V : V \longrightarrow W$ if and only if the composition

$$\mathbb{K}[\mathbb{K}^m] \xrightarrow{\phi} \mathbb{K}[\mathbb{K}^n] \xrightarrow{\rho} \mathbb{K}[V]$$

factors through $\mathbb{K}[W]$. Here ρ is the restriction map.

- (d) For each of the given maps of coordinate rings, describe the corresponding map of algebraic sets. When possible draw pictures. Compare to the corresponding maps on spectra.
- (i) $\mathcal{C}[x, y] \rightarrow \mathcal{C}[t]$ given by $x \mapsto t$ and $y \mapsto t$.
- (ii) $\mathcal{C}[t] \rightarrow \mathcal{C}[x, y]$ given by $t \mapsto x$.
- (iii) $\mathcal{C}[t] \rightarrow \mathcal{C}[x, y, t]/\langle xy - t \rangle$ given by $t \mapsto t$.

Problem 9. Prove that a morphism of algebraic sets is continuous with respect to the Zariski topology.

Problem 10. Show that the image of $\mathbb{K} \rightarrow \mathbb{K}^3$ sending $t \rightarrow (t, t^2, t^3)$ is an algebraic set defined by two polynomials. Verify your answer by constructing the corresponding ring map in *Macaulay2*.

Problem 11. Prove or disprove: The image of every polynomial map $\mathbb{K}^n \rightarrow \mathbb{K}^m$ is an algebraic set. (Warning: this might be field dependent. As a hint, if $\mathbb{K}$ is finite, which subsets of $\mathbb{K}^n$ are algebraic sets?)

You now have the main algebra-geometry dictionary in hand. Over any field $\mathbb{K}$, every algebraic set $V \subset \mathbb{K}^n$ has a coordinate ring $\mathbb{K}[V]$, points of V are the same as $\mathbb{K}$ -algebra maps $\mathbb{K}[V] \rightarrow \mathbb{K}$, and morphisms $V \rightarrow W$ correspond contravariantly to $\mathbb{K}$ -algebra homomorphisms $\mathbb{K}[W] \rightarrow \mathbb{K}[V]$. Thus the assignment

$$V \longmapsto \mathbb{K}[V]$$

always gives a contravariant bridge from geometry to algebra. When $\mathbb{K}$ is algebraically closed, Hilbert's Nullstellensatz says that every finitely generated reduced $\mathbb{K}$ -algebra comes from an algebraic set, so this bridge becomes an anti-equivalence of categories.

Problem 12. Putting It All Together. Assume that $\mathbb{K}$ is algebraically closed. Use the previous parts together with Hilbert's Nullstellensatz II to show that every finitely generated reduced $\mathbb{K}$ -algebra is isomorphic to $\mathbb{K}[V]$ for some algebraic set V . Conclude that the assignment $V \mapsto \mathbb{K}[V]$ defines a (anti-) equivalence from the category of algebraic sets over $\mathbb{K}$ to the category of finitely generated reduced $\mathbb{K}$ -algebras.

Fix a field $\mathbb{K}$. In this worksheet we will explore a special, but important type of rings called valuation rings.

Definition 0.5. A valuation ring of a field $\mathbb{K}$ is a subring $R \subset \mathbb{K}$ with the property that for all $x \in \mathbb{K}^\times$ either $x \in R$ or $x^{-1} \in R$.

Note that, since a valuation ring R of a field $\mathbb{K}$ is a subring of $\mathbb{K}$ it is necessarily a domain. We first study valuation rings directly, then show how certain functions that broadly measure some notion of divisibility, called valuations, produce them. Lastly, we compare discrete and non-discrete examples.

Problem 1. Examples of Valuation Rings. Recall $\mathcal{C}(x, y)$ is the field of formal rational functions in x and y , which concretely is the set of elements f/g where $f, g \in \mathcal{C}[x, y]$ with $g \neq 0$. For each of the following subsets $\mathcal{S} \subset \mathcal{C}(x, y)$ determine whether or not they are valuation rings by showing whether for every $h \in \mathcal{C}(x, y)^\times$ either $h \in \mathcal{S}$ or $h^{-1} \in \mathcal{S}$.

- (a) Hint: Use that $\mathcal{C}[x, y]$ is a UFD to understand when g is divisible by y .

$$\mathcal{S}_1 := \left\{ \frac{f}{g} \mid f, g \in \mathcal{C}[x, y] \text{ and } g \notin \langle y \rangle \right\}$$

- (b) Hint: Consider the element $\frac{x}{y}$.

$$\mathcal{S}_2 := \left\{ \frac{f}{g} \mid f, g \in \mathcal{C}[x, y] \text{ and } g(0, 0) \neq 0 \right\}$$

- (c) Define the *order* of a non-zero polynomial $f \in \mathcal{C}[x, y]$ to be $\mathbf{ord}(f)$, the lowest degree of a monomial in f and $\mathbf{ord}(0) = \infty$.

$$\mathcal{S}_3 := \left\{ \frac{f}{g} \mid f, g \in \mathcal{C}[x, y], g \neq 0, \text{ and } \mathbf{ord}(f) \geq \mathbf{ord}(g) \right\}$$

Problem 2. Ideals in Valuation Rings. Let R be a valuation ring of $\mathbb{K}$.

- (a) Show that for any two non-zero elements $x, y \in R$, either $x|y$ or $y|x$.
- (b) Show that every finitely generated ideal of R is principal.
- (c) Prove that the ideals of R form a totally ordered set. That is, for any two ideals I, J in R , either $I \subseteq J$ or $J \subseteq I$. (Hint: Given $x \in I$ with $x \notin J$ consider whether an element $y \in J$ can divide x .)
- (d) What does the previous parts say about the poset structure on $\text{Spec}(R)$?

Problem 3. Local Rings & Valuation Rings. An arbitrary ring R is said to be a *local ring* if it has exactly one maximal ideal. We often write a local ring R as a pair $(R, \mathfrak{m})$ where $\mathfrak{m} \subset R$ is the unique maximal ideal.

- (a) Describe $\text{mSpec}(R)$ if R is a local ring.
- (b) Describe the closed points of $\text{Spec}(R)$ if R is a local ring.
- (c) Prove that a valuation ring is a local ring.

Problem 4. p -Adic Integers. Fix a prime p . The p -adic numbers $\mathbb{Q}_p$ are the completion of $\mathbb{Q}$ with respect to the p -adic absolute value $|\cdot|_p$, defined as follows: for a nonzero rational number x , write $x = p^k \frac{a}{b}$ where $a, b \in \mathbb{Z}$ are not divisible by p . Then

$$|x|_p = p^{-k}, \quad |0|_p = 0.$$

Note $|\cdot|_p$ extends uniquely to $\mathbb{Q}_p$. The p -adic integers are defined by

$$\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid |x|_p \leq 1\}.$$

Similar to decimal expansion in $\mathbb{R}$, it is a fact that every element $x \in \mathbb{Q}_p$ can be written uniquely as a formal series

$$x = \sum_{i=k}^{\infty} a_i p^i \quad \text{where} \quad a_i \in \{0, 1, \dots, p-1\}, \quad a_k \neq 0,$$

and that

$$\mathbb{Z}_p = \left\{ \sum_{i=0}^{\infty} a_i p^i \mid a_i \in \{0, 1, \dots, p-1\} \right\}.$$

- (a) Prove that the p -adic absolute value is well-defined. (Hint: Use the fact that $\mathbb{Z}$ is a UFD to show that every element $x \in \mathbb{Q}$ can be written uniquely (up to sign and reordering) as $p_1^{a_1} \dots p_t^{a_t}$ where the p_i are distinct primes and $a_i \in \mathbb{Z}$.)
- (b) Prove the p -adic absolute value satisfies the *non-Archimedean (ultrametric) inequality*:

$$|x + y|_p \leq \max\{|x|_p, |y|_p\}.$$

- (c) Prove the p -adic absolute value is multiplicative: $|xy|_p = |x|_p |y|_p$.
- (d) Use parts 2b and 3c to show that $\mathbb{Z}_p$ is closed under addition and multiplication.
- (e) Conclude that $\mathbb{Z}_p$ is a ring (called the ring of p -adic integers).
- (f) Prove that $\mathbb{Z}_p$ is a valuation ring.

The previous exercises suggest why valuation rings are called *valuation* rings: they arise from functions that measure divisibility, generalizing the p -adic absolute value. Such functions are called valuations. An *ordered abelian group* is an abelian group $(\Gamma, +)$ together with a total order $<$ on Γ that is compatible with addition, in the sense that if $\alpha < \beta$ then $\alpha + \gamma < \beta + \gamma$ for all $\alpha, \beta, \gamma \in \Gamma$. We will frequently extend the total ordering on an ordered abelian group Γ to $\Gamma \cup \{\infty\}$ by making the convention that $\gamma < \infty$ for all $\gamma \in \Gamma$. Similarly, we will extend

addition to $\Gamma \cup \{\infty\}$ by declaring $\infty + \gamma = \gamma + \infty = \infty$ for all $\gamma \in \Gamma$.

Definition 0.6. Let $(\Gamma, +, <)$ be an ordered abelian group. A Γ -valued valuation, or just a valuation, on $\mathbb{K}$ is a function: $\mathbf{v} : \mathbb{K} \rightarrow \Gamma \cup \{\infty\}$, which satisfies the following axioms for all $\mathbf{x}, \mathbf{y} \in \mathbb{K}$:

1. $\mathbf{v}(\mathbf{x}\mathbf{y}) = \mathbf{v}(\mathbf{x}) + \mathbf{v}(\mathbf{y})$
2. $\mathbf{v}(\mathbf{x} + \mathbf{y}) \geq \min\{\mathbf{v}(\mathbf{x}), \mathbf{v}(\mathbf{y})\}$, and
3. $\mathbf{v}(\mathbf{x}) = \infty$ if and only if $\mathbf{x} = 0$.

Note that this means $\mathbf{v}$ defines a group homomorphism $\mathbf{v} : \mathbb{K}^\times \rightarrow \Gamma$. Indeed, it is standard to define a valuation as a group homomorphism $\mathbf{v} : \mathbb{K}^\times \rightarrow \Gamma$ satisfying axiom (ii), and then extend it to all of $\mathbb{K}$ by setting $\mathbf{v}(0) = \infty$. This convention is implicit throughout the remainder of the worksheet.

The *value group* of a valuation $\mathbf{v} : \mathbb{K}^\times \rightarrow \Gamma$ is the image of $\mathbf{v}$. A valuation is a *discrete valuation* if its value group is isomorphic to $\mathbb{Z}$. If $\mathcal{F} \subset \mathbb{K}$ is a subfield, we say that $\mathbf{v}$ is a $\mathcal{F}$ -valuation if $\mathbf{v}(\lambda) = 0$ for all $\lambda \in \mathcal{F}^\times$.

Problem 5. Examples of Valuations. Prove that the following are valuations.

(a) The $\mathbf{p}$ -adic valuation:

$$\mathbf{v}_\mathbf{p} : \mathbb{Q}^\times \rightarrow \mathbb{Z}, \quad \mathbf{v}_\mathbf{p}\left(\mathbf{p}^k \frac{\mathbf{a}}{\mathbf{b}}\right) = k$$

where $\mathbf{a}, \mathbf{b} \in \mathbb{Z}$ are not divisible by $\mathbf{p}$.

(b) The order valuation on $\mathcal{C}(\mathbf{x})$:

$$\mathbf{v}_0 : \mathcal{C}(\mathbf{x})^\times \rightarrow \mathbb{Z}, \quad \mathbf{v}_0(\mathbf{f}/\mathbf{g}) = \mathbf{ord}_{\mathbf{x}=0}(\mathbf{f}) - \mathbf{ord}_{\mathbf{x}=0}(\mathbf{g}).$$

(c) More generally, if $\mathbf{a} \in \mathcal{C}$, define

$$\mathbf{v}_\mathbf{a} : \mathcal{C}(\mathbf{x})^\times \rightarrow \mathbb{Z}$$

by order of vanishing at $\mathbf{x} = \mathbf{a}$. Show this is again a valuation.

(d) On $\mathcal{C}(\mathbf{x}, \mathbf{y})$, define

$$\mathbf{v}(\mathbf{f}/\mathbf{g}) = \mathbf{ord}(\mathbf{f}) - \mathbf{ord}(\mathbf{g}),$$

where $\mathbf{ord}(\mathbf{h})$ is the degree of the lowest-degree nonzero term of $\mathbf{h} \in \mathcal{C}[\mathbf{x}, \mathbf{y}]$. Show that this is a valuation with value group $\mathbb{Z}$.

Problem 6. Prove that every ordered abelian group is torsion-free. (Hint: Reduce to the case when $x > 0$.)

Problem 7. Basic Properties of Valuations. Let $v : \mathbb{K}^\times \rightarrow \Gamma$ be an arbitrary valuation. Fix $f, g \in \mathbb{K}^\times$.

- (a) Show that $v(f^{-1}) = -v(f)$.
- (b) Show that if $v(g) < v(f)$, then $v(f+g) = v(g)$. (Hint: Use that $(f+g)+(-f) = g$.)
- (c) Show that $v(f) = v(-f)$. (Hint: Use part 2b.)

Problem 8. Consider the field $\mathcal{C}(x, y)$. Throughout this question π refers to the number $\pi = 3.1415926\dots$

- (a) Show that there is a unique $\mathcal{C}$ -valuation $v_\pi : \mathcal{C}(x, y)^\times \rightarrow (\mathbb{R}, +)$ such that

$$v_\pi(x^a y^b) = a + \pi b.$$

What is the value group of v_π ? (Hint: Use Question 3.)

- (b) Consider the group $\Gamma = \mathbb{Z} \oplus \mathbb{Z}$ ordered by the lexicographical ordering. Show that there exists a unique $\mathcal{C}$ -valuation $v : \mathcal{C}(x, y)^\times \rightarrow \Gamma$ such that $v(x^a y^b) = (a, b)$.
- (c) Are either of these valuations constructed in parts 1a or 2b discrete?

Problem 9. Valuations Determine Valuation Rings. Let v be an arbitrary valuation on $\mathbb{K}$.

- (a) Prove that the set $R \subset \mathbb{K}$ of elements $f \in \mathbb{K}$ with $v(f) \geq 0$ is a valuation ring of $\mathbb{K}$. We call this the *valuation ring* of v . (Hint: Remember our convention about $v(0)$.)
- (b) Show that the following is an ideal of R :

$$\mathfrak{m} := \{f \in \mathbb{K} \mid 0 < v(f)\}.$$

- (c) Prove that if $f \in R$ and $f \notin \mathfrak{m}$ then f is a unit.
- (d) Combine the previous parts to deduce $\mathfrak{m}$ is the unique maximal ideal of R .

- (e) Fix an element $\gamma \in \Gamma$. Prove the sets below are ideals of $\mathbf{R}$:

$$\{f \in \mathbf{R} \mid v(f) > \gamma\} \quad \text{and} \quad \{f \in \mathbf{R} \mid v(f) \geq \gamma\}.$$

Problem 10. Discrete Valuation Rings. Let $v : \mathbb{K}^\times \rightarrow \mathbb{Z}$ be a discrete valuation on $\mathbb{K}$. Let $(\mathbf{R}, \mathbf{m})$ be the corresponding *discrete valuation ring*.

- (a) Let $I \subset \mathbf{R}$ be any non-zero ideal, explain why we can choose an element $f \in I$ such that $v(f)$ is minimal among elements in I .
- (b) Prove that every ideal in $\mathbf{R}$ is principal. In particular, a discrete valuation ring is a PID.
- (c) By part 2b there exists an element $t \in \mathbf{R}$ such that $\langle t \rangle = \mathbf{m}$. Show that every ideal of $\mathbf{R}$ is either 0 or of the form $\langle t^n \rangle$ for some $n \geq 0$.

Problem 11. Consider the valuation v_π on $\mathcal{C}(x, y)$ constructed in Question 1a. Let $(\mathbf{R}, \mathbf{m})$ be the corresponding valuation ring for v_π .

- (a) For a real number $x \in \mathbb{R}_{\geq 0}$ let $\{x\} := x - \lfloor x \rfloor$ denote its fractional part. Fix a positive integer N and consider the sequence consisting of the fractional parts of the first $N + 1$ multiples of π : $\{0\}, \{\pi\}, \{2\pi\}, \dots, \{N\pi\} \in [0, 1)$. Divide the interval $[0, 1)$ into N evenly spaced subintervals:

$$[0, 1) = \left[0, \frac{1}{N}\right) \cup \left[\frac{1}{N}, \frac{2}{N}\right) \cup \dots \cup \left[\frac{N-1}{N}, 1\right).$$

Prove there exist integers $0 \leq k, j \leq N$ such that $\{k\pi\}$ and $\{j\pi\}$ lie in the same subinterval. Deduce that for any positive integer N there exist integers $0 \leq k, j \leq N$ such that $|\{k\pi\} - \{j\pi\}| \leq \frac{1}{N}$. (Hint: Think about pigeons.)

- (b) Using that π is irrational show that $\{j\pi\} \neq \{k\pi\}$ whenever $j \neq k$.
- (c) Prove for every real number $\epsilon > 0$, there exist integers $a, b \in \mathbb{Z}$ such that $0 < a + b\pi < \epsilon$.
- (d) Prove that $\mathbf{m}$ is not finitely generated. (Hint: Use part 3c.)
- (e) Prove that $\mathbf{m}$ is the only non-zero radical ideal of $\mathbf{R}$. (Hint: If $I \subset \mathbf{R}$ is a non-zero radical ideal, let $f \in I$ be non-zero. For an arbitrary element $h \in \mathbf{m}$ show that $h^n \in \langle f \rangle$ for $n \gg 0$.)
- (f) Describe $\text{Spec}(\mathbf{V})$.

Problem 12. Consider the valuation ν on $\mathcal{C}(x, y)$ constructed in Question 2b. Let $(R, \mathfrak{m})$ be the corresponding valuation ring for ν .

(a) Show that the maximal ideal of R is

$$\mathfrak{m} = \{f \in \mathcal{C}(x, y) \mid \nu(f) \geq (0, 1)\} = \langle y \rangle.$$

(b) Show that the ideal $\langle x \rangle \subset R$ is not prime. (Hint: $x = xy^{-1} \cdot y$.)

(c) Show that the set below is a prime, non-maximal ideal of R :

$$P := \{f \in R \mid \nu(f) = (a, b) \text{ where } a \geq 1\} \cup \{0\}.$$

(d) Use the valuation to construct an infinite strictly ascending chain of ideals in R :

$$\langle xy^{-1} \rangle \subset \langle xy^{-2} \rangle \subset \langle xy^{-3} \rangle \subset \cdots.$$

Explain why all inclusions are strict.

Problem 13. Valuation Rings Determine Valuations. Let R be a valuation ring of $\mathbb{K}$ with $R \neq \mathbb{K}$. This problem shows how to recover a valuation from the divisibility structure of R .

(a) Let Γ' be the set of nonzero principal ideals of R :

$$\Gamma' = \{\langle a \rangle \subset R \mid a \in R, a \neq 0\}.$$

Define a multiplication on Γ' by ideal multiplication: $\langle a \rangle \cdot \langle b \rangle = \langle ab \rangle$.

(a) Show that this operation is well-defined and makes Γ' into an abelian monoid (associative, commutative, binary operation with an identity element).

(b) Show that Γ' is totally ordered by inclusion: for distinct elements $\langle a \rangle, \langle b \rangle \in \Gamma'$, either $\langle a \rangle \subseteq \langle b \rangle$ or $\langle b \rangle \subseteq \langle a \rangle$.

(b) Define an equivalence relation $\sim$ on $\mathbb{K}^\times$ by $x \sim y$ if and only if $x = uy$ for some unit $u \in R^\times$. Let $\Gamma = \mathbb{K}^\times / \sim$ denote the set of equivalence classes. Given $x \in \mathbb{K}^\times$ write $[x]$ its equivalence class in Γ .

(a) Show that $\sim$ is a well-defined equivalence relation.

(b) Show that multiplication in $\mathbb{K}^\times$ induces a well-defined operation on Γ , making it into an abelian group.

- (c) Define $[x] \leq [y]$ if and only if $yx^{-1} \in R$. Show this is a well-defined total order on Γ .
- (d) Deduce that Γ is an ordered abelian group.
- (e) Show that the map $\Gamma' \rightarrow \Gamma$ given by $\langle a \rangle \mapsto [a]$ is an order-preserving embedding whose image consists exactly of the elements $\geq [1]$.
- (c) Define $\nu : \mathbb{K}^\times \rightarrow \Gamma$ by $\nu(x) = [x]$. Prove that ν is a valuation on $\mathbb{K}$ with value group Γ and

$$R = \{x \in \mathbb{K} \mid \nu(x) \geq [1]\}.$$

Moreover, $\nu(x) \in \Gamma'$ if and only if $x \in R$.

Worksheet 3.1: Gröbner Bases Pt. II

Fix a field $\mathbb{K}$. Unless otherwise specified the ring throughout this worksheet will be $\mathbb{K}[x_1, \dots, x_n]$. In the previous worksheet we introduced monomial orderings and leading terms. In this worksheet we return to the multivariable division algorithm, and then use it to study monomial ideals, initial ideals, and Gröbner bases. The main goal is to push the story all the way to a major finiteness theorem: every ideal of $\mathbb{K}[x_1, \dots, x_n]$ is finitely generated.

Recall a *monomial ordering* on $\mathbb{K}[x_1, \dots, x_n]$ is a total well-ordering $<$ on $\mathbb{Z}_{\geq 0}^n$ such that if $\alpha < \beta$ then $\alpha + \gamma < \beta + \gamma$ for all $\gamma \in \mathbb{Z}_{\geq 0}^n$. Using the bijection between monomials in $\mathbb{K}[x_1, \dots, x_n]$ and elements of $\mathbb{Z}_{\geq 0}^n$ we think of a monomial ordering as giving us a way to compare monomials by saying $x^\alpha < x^\beta$ if and only if $\alpha < \beta$. In Worksheet 1.3 we saw several important monomial orders: lexicographic (lex) order, graded lexicographic (grlex) order, and graded reverse lexicographic (grevlex) order. Please review that worksheet for their definitions.

Returning to our goal of having a division algorithm in $\mathbb{K}[x_1, \dots, x_n]$ we are faced with two challenges compared to the familiar setting of $\mathbb{K}[x]$:

- (i) In several variables the notion of “leading term” is ambiguous. In one variable, the leading term is simply the term of highest degree. In several variables this no longer suffices: for instance, what should the leading term of $x^2 + xy + y^2$ be? All three terms have total degree 2.
- (ii) We want a division algorithm that can answer questions about ideals, i.e., membership, intersections, sums, products, radicals. In $\mathbb{K}[x]$ every ideal is principal, so dividing by a single generator is enough. For $n > 1$, however, $\mathbb{K}[x_1, \dots, x_n]$ is no longer a PID, so ideals generally require multiple generators. We therefore need a way to divide a polynomial by an entire *list* of polynomials.

As seen on Worksheet 1.3, once a monomial ordering has been fixed, we give a meaningful definition of leading term, leading monomial, and leading coefficient; solving issue (i). Note while we normally write $LT(f)$, $LM(f)$ and $LC(f)$ for these quantities; they do depend on the chosen order $<$, and so would more properly be called $LT_{<}(f)$, $LM_{<}(f)$ and $LC_{<}(f)$ to highlight this fact.

To resolve issue (ii), we generalize long division as follows. In single-variable polynomial long division, we repeatedly ask: “Does the leading term of the divisor divide the current leading term of the dividend?” If yes, we subtract off the appropriate multiple; if not, we are done. The multivariable algorithm works the same way, except that instead of checking a single divisor we scan through an *ordered* list of divisors $G = (g_1, \dots, g_s)$ and use the first one whose leading term divides the current leading term. More precisely, the algorithm maintains a “working polynomial” p (initialized to f), quotients $q_1, \dots, q_s$ (initialized to 0), and a remainder r (initialized to 0), then repeats:

- **Try to Divide.** Walk through the list $g_1, g_2, \dots, g_s$ in order. If you find some g_i whose leading term $LT(g_i)$ divides $LT(p)$, subtract the appropriate multiple: update $q_i \leftarrow q_i + LT(p)/LT(g_i)$ and $p \leftarrow p - (LT(p)/LT(g_i)) g_i$. Then return to the start of the list and try again with the new p .
- **Move to Remainder.** If *no* g_i in the list has a leading term dividing $LT(p)$, then the current leading term of p cannot be reduced further. Move it to the remainder: $r \leftarrow r + LT(p)$ and $p \leftarrow p - LT(p)$.

We continue until the working polynomial p is equal to 0. Proving this algorithm terminates as expected, which you will do later in this worksheet, amounts essentially to proving the following theorem

Theorem 0.7. *Fix a monomial ordering $<$ on $\mathbb{K}[x_1, \dots, x_n]$ and let $G = (g_1, \dots, g_s)$ be an ordered list of nonzero polynomials. If $f \in \mathbb{K}[x_1, \dots, x_n]$ then there exist $q_1, \dots, q_s, r \in \mathbb{K}[x_1, \dots, x_n]$ such that*

$$f = q_1 g_1 + \dots + q_s g_s + r,$$

where no monomial appearing in r is divisible by any of $LM(g_1), \dots, LM(g_s)$.

Written in pseudocode the algorithm for multivariable polynomial division is shown below.

Algorithm 2 Division by an ordered list $G = (g_1, \dots, g_s)$

Require: $f, g_1, \dots, g_s$ **Ensure:** $q_1, \dots, q_s, r$ $q_1, \dots, q_s \leftarrow 0; r \leftarrow 0; p \leftarrow f$ **while** $p \neq 0$ **do** \leftarrow 0 **for** $i = 1, \dots, s$ **do** **if** $\text{LT}(g_i)$ divides $\text{LT}(p)$ **then** $q_i \leftarrow q_i + \text{LT}(p)/\text{LT}(g_i)$ $p \leftarrow p - (\text{LT}(p)/\text{LT}(g_i))g_i$ divides $\leftarrow 1$ **break** **end if** **end for** **if** divides = 0 **then** $r \leftarrow r + \text{LT}(p)$ $p \leftarrow p - \text{LT}(p)$ **end if****end while**

Problem 1. Review of Monomial Orders.Work in $\mathbb{K}[x, y, z]$ and assume $x > y > z$.

- (a) Order the monomials $x^2y, xy^2, xz^2, y^3, yz^4$ from largest to smallest with respect to lex, graded lex, and graded reverse lex.
- (b) Compute $\text{LM}(f)$, $\text{LC}(f)$, and $\text{LT}(f)$ for $f = 4x^2y - 3xy^2 + y^4 + 2z^5$ with respect to lex and grevlex.
- (c) Prove that **1** is the smallest monomial for every monomial ordering.
- (d) Deduce that every strictly decreasing sequence of monomials terminates.
- (e) Explain why part (d) should make you optimistic that a division algorithm might terminate.

Problem 2. Examples of Multivariable Division.For this question, we will work in $\mathbb{K}[x, y]$ considered with the lex order where $x > y$.

- (a) Divide $f = x^2y + xy + y - x - 2$ by the ordered list $G = (xy - 1, y - 1)$. Record

the successive values of $\mathbf{p}$, the quotients, and the remainder after each update.

- (b) Divide $f = xy^2 - x$ by the ordered list $G = (xy - 1, y^2 - 1)$. Record the successive values of $\mathbf{p}$, the quotients, and the remainder after each update.
- (c) Repeat (b), but with the ordered list $(y^2 - 1, xy - 1)$, i.e. where the order of G is swapped.
- (d) Show directly that $xy^2 - x \in \langle xy - 1, y^2 - 1 \rangle$.
- (e) What do your computations show about division in several variables? In particular, what is always true when the remainder is 0, and what can go wrong when the remainder is non-zero?

Problem 3. Termination of Division Algorithm

Let $G = (g_1, \dots, g_s)$ be an ordered list of non-zero polynomials.

- (a) Show that throughout the algorithm one always has $f = q_1g_1 + \dots + q_sg_s + p + r$.
- (b) Show that after each pass through the `while` loop, the leading monomial of $\mathbf{p}$ strictly decreases with respect to the given monomial order.
- (c) Use Question 1(d) to prove that the algorithm terminates.
- (d) Prove that on termination the output satisfies $f = q_1g_1 + \dots + q_sg_s + r$, and no monomial appearing in r is divisible by any of $\text{LM}(g_1), \dots, \text{LM}(g_s)$.
- (e) Explain why a zero remainder certifies ideal membership: if dividing f by G produces remainder 0, then $f \in \langle g_1, \dots, g_s \rangle$.

Question 2 reveals an important new feature of polynomial division in several variables. Unlike the one-variable case, the remainder obtained by dividing a polynomial f by polynomials $g_1, \dots, g_t$ depends on two choices: i) the order of our list of divisors and ii) the monomial order we have fixed. Because of this, division does *not* immediately solve the ideal membership problem in several variables. In other words, even if dividing f by the list $(g_1, \dots, g_t)$ produces a nonzero remainder, it may still happen that $f \in \langle g_1, \dots, g_t \rangle$. So, in several variables, a nonzero remainder does *not* by itself show that $f \notin \langle g_1, \dots, g_t \rangle$. We will eventually develop a way to fix this problem. Before we do, however, it is useful to study a special class of ideals for which ideal membership *is* controlled purely by divisibility of monomials.

Definition 0.8. An ideal $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ is a monomial ideal if it can be generated by monomials.

We shall see shortly that a monomial ideal is entirely determined by the set of monomials belonging to the ideal. Since monomials in $\mathbb{K}[x_1, \dots, x_n]$ are in bijection with $\mathbb{Z}_{\geq 0}^n$ this means we can – and often do think of monomial ideals as corresponding to subsets of $\mathbb{Z}_{\geq 0}^n$. With this viewpoint, the following classical result guarantees that monomial ideals are well-behaved.

Theorem 0.9 (Dickson’s Lemma). *Every subset of $\mathbb{Z}_{\geq 0}^n$ has only finitely many minimal elements with respect to the coordinatewise order $\leq$.*

Motivated by this theorem we say a *minimal monomial generating set* for a monomial ideal I is a finite set of monomials $\{m_1, \dots, m_t\}$ such that $I = \langle m_1, \dots, m_t \rangle$ and no m_i divides m_j for $i \neq j$. Equivalently, no generator is redundant: removing any m_i from the set yields a strictly smaller ideal.

Problem 4. Properties of Monomial Ideals.

- (a) Decide which of the following ideals are monomial ideals:

$$\langle x^2, xy^3, z \rangle, \quad \langle x + y, y^2 \rangle, \quad \langle x^2 - y, y^3 \rangle, \quad \langle x^2, xy, y^4 + x \rangle.$$

- (b) Let $m_1, \dots, m_t$ and m be monomials. Prove that $m \in \langle m_1, \dots, m_t \rangle$ if and only if $m_i \mid m$ for some i .
- (c) Deduce that a polynomial $f \in \mathbb{K}[x_1, \dots, x_n]$ lies in I if and only if every monomial appearing in f is divisible by one of $m_1, \dots, m_t$.
- (d) Find a minimal monomial generating set for each of the following ideals:

$$J = \langle x^3, x^2y, xy^2, y^4, x^4y, x^2y^3 \rangle \quad \text{and} \quad L = \langle x^2z, xyz, y^2z, z^3, xz^4 \rangle.$$

Draw a picture of the subset of $\mathbb{Z}_{\geq 0}^2$ corresponding to the monomials in J .

Problem 5. Understanding Dickson’s Lemma.

- (a) Let $A \subseteq \mathbb{Z}_{\geq 0}^n$ and set $I_A := \langle x^\alpha \mid \alpha \in A \rangle$. If $A_{\min}$ denotes the set of minimal elements of A , show that every monomial in I_A is divisible by some x^α with $\alpha \in A_{\min}$.
- (b) Explain why the statement “every subset of $\mathbb{Z}_{\geq 0}^n$ has finitely many minimal elements with respect to $\leq$ ” is equivalent to the statement “every monomial ideal in $\mathbb{K}[x_1, \dots, x_n]$ is finitely generated.”
- (c) Restate Dickson’s Lemma entirely in terms of monomial ideals in $\mathbb{K}[x_1, \dots, x_n]$.

- (d) Prove your monomial ideal version of Dickson's Lemma in the case when $n = 1$.

Problem 6. Proving Dickson's Lemma. We now prove the monomial ideal version of Dickson's Lemma by induction on the number of variables. Your work in Question 4d serves as the base case. Assume Dickson's Lemma holds in $n - 1$ variables. Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be a monomial ideal. For $k \geq 0$ let

$$J_k = \left\{ f \in \mathbb{K}[x_1, \dots, x_{n-1}] \mid fx_n^k \in I \right\} \cup \{0\}.$$

- (a) Prove that J_k is a monomial ideal in $\mathbb{K}[x_1, \dots, x_{n-1}]$ for all $k \geq 0$.
(b) Show that the J_k form an ascending chain.

$$J_0 \subseteq J_1 \subseteq J_2 \subseteq \dots$$

(Hint: What happens when you multiply something in J_k by x_n ?)

- (c) Let $L = \bigcup_{d \geq 0} J_d$. Show that L is a monomial ideal in $\mathbb{K}[x_1, \dots, x_{n-1}]$.
(d) Prove that there exists $N \in \mathbb{Z}_{\geq 0}$ such that $L = J_k$ for all $k \geq N$. (Hint: Use the inductive hypothesis to show that L is finitely generated.)
(e) For each $0 \leq k \leq N$, choose monomial generators $m_{k,1}, \dots, m_{k,r_k}$ for J_k . Via the inclusion $\mathbb{K}[x_1, \dots, x_{n-1}] \subset \mathbb{K}[x_1, \dots, x_n]$ view these as elements in $\mathbb{K}[x_1, \dots, x_n]$. Prove that I is generated by the finite set

$$\{m_{k,i}x_n^k \mid 0 \leq k \leq N, 1 \leq i \leq r_k\}.$$

- (f) Conclude Dickson's Lemma for all n .

Problem 7. Minimal monomial generators.

Let I be a monomial ideal, and let $M(I)$ be the set of monomials in I which are minimal under divisibility.

- (a) Use Dickson's Lemma to show that $M(I)$ is finite.
(b) Prove that $I = \langle M(I) \rangle$.
(c) Prove that every monomial generating set of I contains $M(I)$. Conclude that $M(I)$ is the unique minimal monomial generating set of I .

Monomial ideals are easier to study because membership is determined entirely by divisibility of monomials, and Dickson's Lemma gives strong finiteness properties. For a general ideal, we would like to recover similar structures by extracting the leading term of each polynomial. This leads to the following.

Definition 0.10. Let $<$ be a monomial order on $\mathbb{K}[x_1, \dots, x_n]$. The initial ideal of a nonzero ideal $I \subset \mathbb{K}[x_1, \dots, x_n]$ with respect to $<$ is:

$$\text{in}_{<}(I) := \langle LT(f) \mid f \in I \rangle.$$

A finite subset $\mathcal{G} = \{g_1, \dots, g_s\} \subset I$ is a Gröbner basis for I with respect to $<$ if $\text{in}_{<}(I) = \langle LT(g_1), \dots, LT(g_s) \rangle$.

It is standard to set $\text{in}_{<}(\langle 0 \rangle) = \langle 0 \rangle$. Note a priori there is no reason for Gröbner bases to exist in general or be interesting. In the next few exercises you will show that Dickson's Lemma not only implies that Gröbner bases exist, but also they will give an answer to the ideal membership problem.

Problem 8. Initial Ideals of Monomial Ideals.

Let $I \subset \mathbb{K}[x_1, \dots, x_n]$ be a monomial ideal. Prove that $\text{in}_{<}(I) = I$ for every monomial order $<$.

Problem 9.

For this problem consider $\mathbb{K}[x, y]$ with the lex order where $x > y$. Let $I = \langle xy - 1, y^2 - 1 \rangle$.

- (a) Find explicit generators for the ideal generated by the leading terms of $xy - 1$ and $y^2 - 1$.
- (b) Prove that $\{xy - 1, y^2 - 1\}$ is not a Gröbner basis for I by finding an element in I not contained in the ideal computed in 1a. (Hint: Consider the element $x - y$.)
- (c) Use *Macaulay2* to find a Gröbner basis for I .

Problem 10. Existence of Gröbner Bases.

Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be an ideal and $<$ be a monomial order.

- (a) Show that $\text{in}_{<}(I)$ is a monomial ideal.
- (b) By Dickson's Lemma, $\text{in}_{<}(I) = \langle m_1, \dots, m_t \rangle$ for a minimal monomial generating set $m_1, \dots, m_t$. Show that there exist elements $g_1, \dots, g_t \in I$ such that $LT(g_i)$ divides m_i for all i .

- (c) Show that the minimality of $\mathbf{m}_1, \dots, \mathbf{m}_t$ implies $LT(\mathbf{g}_i) = c_i \mathbf{m}_i$ for some scalar $c_i \in \mathbb{K}^\times$
- (d) Conclude that $\mathcal{G} = \{\mathbf{g}_1, \dots, \mathbf{g}_t\}$ is a Gröbner basis of I .

Problem 11. Ideal Membership & Gröbner Bases.

Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be an ideal and $<$ be a monomial order. Fix a Gröbner basis $\mathcal{G} = (\mathbf{g}_1, \dots, \mathbf{g}_t)$ for I , i.e. elements $\mathbf{g}_i \in I$ such that $\mathbf{m}_i := LT(\mathbf{g}_i)$ generate $\text{in}_<(I)$.

- (a) Given $f \in \mathbb{K}[x_1, \dots, x_n]$ consider the division of f by $\mathcal{G} = (\mathbf{g}_1, \dots, \mathbf{g}_t)$ obtaining

$$f = q_1 \mathbf{g}_1 + \dots + q_t \mathbf{g}_t + r,$$

as in Theorem 0.7. Show that $f \in I$ if and only if $r \in I$.

- (b) Assume $f \in I$ and $r \neq 0$. Show that $LT(r) \in \text{in}_<(I)$, and conclude $LT(r)$ is divisible by one of $\mathbf{m}_1, \dots, \mathbf{m}_t$. Why does this contradict the defining property of the remainder? Conclude that every $f \in I$ has remainder 0 on division by $\mathcal{G}$.
- (c) Membership Test. Prove that if $\mathcal{G}$ is a Gröbner basis for I , then for any polynomial $f \in \mathbb{K}[x_1, \dots, x_n]$

$$f \in I \iff \text{the remainder on dividing } f \text{ by } \mathcal{G} \text{ is } 0.$$

- (d) Prove that $I = \langle \mathbf{g}_1, \dots, \mathbf{g}_t \rangle$. (Hint: The inclusion $\langle \mathbf{g}_1, \dots, \mathbf{g}_t \rangle \subset I$ should be clear. For the other, let $f \in I$ then apply the division algorithm and the membership test.)

Notice that Questions 10 and 11 together show; given any ideal $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ and any monomial order $<$, a Gröbner basis for I exists (Question 10), and every Gröbner basis generates I (Question 11(d)). Since a Gröbner basis is finite by construction, every ideal of $\mathbb{K}[x_1, \dots, x_n]$ is finitely generated. This naturally leads us to our first encounter with the following important finiteness condition.

Definition 0.11. A ring R is Noetherian if and only if every ascending chain of ideals:

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$$

stabilizes in the sense that there exists $N \in \mathbb{Z}_{\geq 1}$ such that $I_n = I_{n+1}$ for all $n \geq N$.

It is common to say a ring is Noetherian if it has the ACC property where ACC is short for ascending chain condition. You will show below a ring R being Noetherian is equivalent

to every ideal of R being finitely generated. Thus, you have proven that $\mathbb{K}[x_1, \dots, x_n]$ is Noetherian. Since a field $\mathbb{K}$ is Noetherian, you have actually proven a special, but important case of the following theorem.

Theorem 0.12 (Hilbert's Basis Theorem). *Let R be a ring. If R is Noetherian, then $R[x]$ is Noetherian.*

The remainder of this worksheet is devoted to proving the equivalence between the ACC and finite generation of ideals, and proving Hilbert's basis theorem in general.

Problem 12. Properties of Noetherian Rings.

For this question let R be any commutative ring, and consider an ascending chain of ideals in R :

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots.$$

- (a) Show that $J = \bigcup_{k \geq 1} I_k$ is an ideal of R .
- (b) If J is finitely generated, say $J = \langle a_1, \dots, a_t \rangle$, explain why there exists an integer $N \in \mathbb{Z}_{\geq 1}$ such that $a_1, \dots, a_t \in I_N$. Conclude that the chain of ideals stabilizes.
- (c) Explain why you just showed that if every ideal of R is finitely generated then R is Noetherian.
- (d) Let I be an ideal of R and suppose every ascending chain of ideals in R stabilizes. Pick any $a_1 \in I$. If $\langle a_1 \rangle \neq I$, pick $a_2 \in I \setminus \langle a_1 \rangle$, obtaining $\langle a_1 \rangle \subsetneq \langle a_1, a_2 \rangle$. Continue in this way. Use the ascending chain condition to show this process must terminate, and conclude that I is finitely generated.
- (e) Show that if R is Noetherian and $I \subseteq R$ is an ideal, then R/I is Noetherian.

Problem 13. Hilbert's Basis Theorem.

Let R be a Noetherian ring, and let $I \subseteq R[x]$ be an ideal. For $k \in \mathbb{Z}_{\geq 0}$, define

$$J_k := \left\{ a \in R \mid \begin{array}{l} \text{there exists } f \in I \\ \text{LT}(f) = ax^k \end{array} \right\} \cup \{0\}.$$

Equivalently, $a \in J_k$ if and only if there exists $f \in I$ with $\text{LT}(f) = ax^k$ (or $a = 0$). Thus J_k is the set of all leading coefficients of polynomials of degree k in I , together with 0. Note that we are viewing $R[x]$ as a polynomial ring in the single variable x over R . We define the leading term of a polynomial $f = a_k x^k + a_{k-1} x^{k-1} + \cdots + a_1 x + a_0$ (with $a_k \neq 0$) to be $\text{LT}(f) = a_k x^k$.

- (a) Show that J_k is an ideal of R for all $k \geq 0$.

- (b) Show that the J_k form an ascending chain $J_0 \subseteq J_1 \subseteq J_2 \subseteq \cdots$.
- (c) Let $L = \bigcup_{d \geq 0} J_d$. Show that L is an ideal in R .
- (d) Prove that there exists $N \in \mathbb{Z}_{\geq 0}$ such that $L = J_k$ for all $k \geq N$. (Hint: Use that R is Noetherian.)

For each $0 \leq k \leq N$, choose generators $\mathbf{a}_{k,1}, \dots, \mathbf{a}_{k,r_k}$ for J_k and polynomials $f_{k,i} \in I$ such that $\text{LT}(f_{k,i}) = \mathbf{a}_{k,i}x^k$. We will now show that I is equal to

$$M := \langle f_{k,i} \mid 0 \leq k \leq N, 1 \leq i \leq r_k \rangle.$$

The inclusion $M \subset I$ is clear. We prove the reverse inclusion $I \subseteq M$ by induction on degree. For the base case suppose $g \in I$ and $\deg(g) = 0$. Since $g \in I$ and $\text{LT}(g) = cx^0 = c$, we have $c \in J_0$. Using that $J_0 = \langle \mathbf{a}_{0,1}, \dots, \mathbf{a}_{0,r_0} \rangle$ we may write c as $c = b_1 \mathbf{a}_{0,1} + \cdots + b_{r_0} \mathbf{a}_{0,r_0}$ with $b_i \in R$. For each i we have $\text{LT}(f_{0,i}) = \mathbf{a}_{0,i}x^0 = \mathbf{a}_{0,i}$ so $f_{0,i} = \mathbf{a}_{0,i}$. Hence $\mathbf{a}_{0,i}$ is in M for all i implying $g \in M$ proving the base case.

Now suppose every element of I with degree less than d belongs to M . Let $g \in I$ with $\deg(g) = d$ and $\text{LT}(g) = cx^d$. Our goal is to find $h \in M$ with $\text{LT}(h) = cx^d$, so that $g - h \in I$ has degree less than d and the inductive hypothesis applies. Set $n = \min\{d, N\}$. Since the ascending chain $J_0 \subseteq J_1 \subseteq \cdots$ stabilizes at J_N , we have $J_d = J_n$, so $c \in J_n = \langle \mathbf{a}_{n,1}, \dots, \mathbf{a}_{n,r_n} \rangle$. Write

$$c = b_1 \mathbf{a}_{n,1} + \cdots + b_{r_n} \mathbf{a}_{n,r_n}$$

for some $b_i \in R$, and define

$$h = \sum_{i=1}^{r_n} b_i x^{d-n} f_{n,i}.$$

(Note: if $d \leq N$ then $n = \min\{d, N\} = d$ and the $x^{d-n} = 1$.) Each $f_{n,i} \in M$, so $h \in M$.

- (a) Show that $\text{LT}(h) = cx^d$. From this deduce that $g - h \in I$ and either $\deg(g - h) < d$ or $g - h = 0$.
- (b) Using the inductive hypothesis conclude that $g - h \in M$, and show this implies $g \in M$. Conclude that $I \subset M$. Explain why this concludes the proof of Hilbert's Basis Theorem.
- (c) Prove via induction on n that if R is Noetherian then $R[x_1, \dots, x_n]$ is Noetherian.
- (d) Compare this proof to the proof of Dickson's Lemma we gave above.

This worksheet has introduced two important concepts that will recur throughout the

course: the usefulness of Gröbner bases and the importance of finiteness conditions such as Noetherianity. A significant open thread raised by this worksheet is that while we know theoretically that Gröbner bases exist, we do not yet know how to actually compute them. We will return to this question later.

It is fitting that Gröbner bases and Noetherianity are so linked. Hilbert's original proof of his Basis Theorem was non-constructive, even over $\mathbb{K}[x_1, \dots, x_n]$, which – as the story goes – came as a shock to many mathematicians of the time. Paul Gordan, a prominent algebraist, reportedly remarked, perhaps seriously, perhaps in jest, “*This is not mathematics. This is theology.*” (Gordan later gave another proof of the theorem, one which in a sense also proved Dickson's Lemma some 10 years before Dickson.) On the other hand, Gröbner bases have come to serve as the foundational tool of computational commutative algebra, allowing us to compute and make explicit a great many things that Hilbert's methods could not.
